

INMOTC 2026 (MP region)

ALGEBRA

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§1 Polynomials

For further problems, we refer to [Goy21].

§1.1 Warm up

Example 1.1. Factorize the polynomial $x^8 + x^4 + 1$ into factors of at most the second degree.

Summary — Expressing an expression as a difference of two squares yields a factorization.

Solution 1. Note that

$$\begin{aligned}
 x^8 + x^4 + 1 &= x^8 + 2x^4 + 1 - x^4 \\
 &= (x^4 + 1)^2 - (x^2)^2 \\
 &= (x^4 - x^2 + 1)(x^4 + 1 + x^2) \\
 &= (x^4 + 2x^2 + 1 - 3x^2)(x^4 + 2x^2 + 1 - x^2) \\
 &= ((x^2 + 1)^2 - (\sqrt{3}x)^2)((x^2 + 1)^2 - x^2) \\
 &= (x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1)(x^2 - x + 1)(x^2 + x + 1).
 \end{aligned}$$

■

Example 1.2. Show that

$$2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4 = (a+b+c)(a+b-c)(b+c-a)(c+a-b).$$

Summary — Complete the squares.

Walkthrough —

- (a) Try to see what would happen if we were allowed to change the signs!
 (b) Change the signs and consider

$$a^4 + b^4 + c^4 - 2a^2b^2 - 2b^2c^2 - 2c^2a^2.$$

This is “almost” $(a^2 - b^2 - c^2)^2$!! To be precise

$$(a^2 - b^2 - c^2)^2 = a^4 + b^4 + c^4 - 2a^2b^2 + 2b^2c^2 - 2c^2a^2.$$

- (c) Let us continue with the above, and write $2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4$ in terms $(a^2 - b^2 - c^2)^2$ as follows.

$$\begin{aligned}
 2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4 \\
 = 4b^2c^2 - (a^4 + b^4 + c^4 - 2a^2b^2 + 2b^2c^2 - 2c^2a^2).
 \end{aligned}$$

- (d) Does the above help?

Solution 2. Note that

$$2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4$$

$$\begin{aligned}
&= 4b^2c^2 - (a^4 + b^4 + c^4 - 2a^2b^2 + 2b^2c^2 - 2c^2a^2) \\
&= (2bc)^2 - (a^2 - b^2 - c^2)^2 \\
&= (2bc - (a^2 - b^2 - c^2))(2bc + a^2 - b^2 - c^2) \\
&= (2bc + b^2 + c^2 - a^2)(a^2 - (b^2 + c^2 - 2bc)) \\
&= ((b + c)^2 - a^2)(a^2 - (b - c)^2) \\
&= (a + b + c)(b + c - a)(a + b - c)(a - b + c) \\
&= (a + b + c)(a + b - c)(b + c - a)(c + a - b).
\end{aligned}$$

■

Solution 3. Write¹

$$\begin{aligned}
a &= y + z, \\
b &= z + x, \\
c &= x + y.
\end{aligned}$$

Note that

$$\begin{aligned}
&2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4 \\
&= 2(y + z)^2(z + x)^2 + 2(z + x)^2(x + y)^2 + 2(x + y)^2(y + z)^2 \\
&\quad - (y + z)^4 - (z + x)^4 - (x + y)^4 \\
&= 2(z^2 + 2yz + y^2)(z^2 + 2zx + x^2) \\
&\quad + 2(x^2 + 2zx + z^2)(x^2 + 2xy + y^2) \\
&\quad + 2(y^2 + 2xy + x^2)(y^2 + 2yz + z^2) \\
&\quad - (y + z)^4 - (z + x)^4 - (x + y)^4 \\
&= 2 \sum_{\text{cyc}} (x^4 + x^2(y^2 + z^2 + 2x(y + z)) + yz(2x + y)(2x + z)) - \sum_{\text{cyc}} (x + y)^4 \\
&= 2 \sum_{\text{cyc}} (x^4 + x^2y^2 + z^2x^2 + 2x^3(y + z) + 4x^2yz + 2xyz(y + z) + y^2z^2) \\
&\quad - \sum_{\text{cyc}} (x + y)^4 \\
&= 2(x^4 + y^4 + z^4) + 6(x^2y^2 + y^2z^2 + z^2x^2) + 16xyz(x + y + z) + 4 \sum_{\text{cyc}} x^3(y + z)
\end{aligned}$$

¹It is good ask the following simple and innocent question: how can one **write** a, b, c as stated above? Does it mean that given any three real numbers a, b, c , one can find real numbers x, y, z such that $a = y + z, b = z + x, c = x + y$? A crucial point to note is that one very often deals with indeterminates (aka variables) instead of real numbers. In the above, a, b, c could be indeterminates instead of being real numbers! What do we do in that case? Is it so that there are **indeterminates** x, y, z such that the six indeterminates a, b, c, x, y, z satisfy $a = y + z, b = z + x, c = x + y$? As of now, **let's not worry about any of these!**. Just keep in mind the message that at times, **we need to be quite careful about what we do!**

$$\begin{aligned}
& - \sum_{\text{cyc}} (x+y)^4 \\
& = 2(x^4 + y^4 + z^4) + 6(x^2y^2 + y^2z^2 + z^2x^2) + 16xyz(x+y+z) + 4 \sum_{\text{cyc}} x^3(y+z) \\
& - \sum_{\text{cyc}} (x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4) \\
& = 16xyz(x+y+z) \\
& = (a+b+c)(a+b-c)(b+c-a)(c+a-b).
\end{aligned}$$

■

Remark. Please go through the footnote next to the word **Write** from the above solution. This footnote would explain that just writing **Write** $a = y + z, b = z + x, c = x + y$ requires more care! To address this issue, replace **Write** $a = y + z, b = z + x, c = x + y$ in the above solution by the following.

Consider the real numbers x, y, z defined by

$$\begin{aligned}
x &= \frac{1}{2}(b+c-a), \\
y &= \frac{1}{2}(c+a-b), \\
z &= \frac{1}{2}(a+b-c).
\end{aligned}$$

Note that

$$a = y + z, b = z + x, c = x + y$$

holds.

Example 1.3. Find numbers a, b, c, d for which the equation

$$\frac{2x-7}{4x^2+16x+15} = \frac{a}{x+c} + \frac{b}{x+d}$$

would be an identity.

Walkthrough — Factorize the denominator into linear factors. Then expressing the numerator as a linear combination of those factors would provide such an identity.

Solution 4. Note that

$$\frac{2x-7}{4x^2+16x+15} = \frac{2x-7}{(2x+3)(2x+5)}.$$

Hence, if $2x-7$ can be expressed as

$$p(2x+3) + q(2x+5),$$

then $\frac{2x-7}{4x^2+16x+15}$ can be expressed as a sum of two fractions, each having a constant in the numerator and a linear polynomial in the denominator.

One way to find if there are any such p, q , is to assume first that there are such real numbers p and q such that

$$2x - 7 = p(2x + 3) + q(2x + 5)$$

holds². Substituting $x = -\frac{5}{2}$, we obtain $-2p = -12$, which gives $p = 6$. Next, substituting $x = -\frac{3}{2}$, we obtain $2q = -10$, which implies $q = -5$.

Note that

$$6(2x + 3) + (-5)(2x + 5) = 12x + 18 - 10x - 25 = 2x - 7$$

holds³. Using it, we obtain

$$\begin{aligned} \frac{2x-7}{4x^2+16x+15} &= \frac{2x-7}{(2x+3)(2x+5)} \\ &= \frac{6(2x+3) + (-5)(2x+5)}{(2x+3)(2x+5)} \\ &= \frac{6}{2x+5} - \frac{5}{2x+3} \\ &= \frac{3}{x+\frac{5}{2}} - \frac{\frac{5}{2}}{x+\frac{3}{2}}. \end{aligned}$$

Hence, we may take

$$a = 3, b = -\frac{5}{2}, c = \frac{5}{2}, d = \frac{3}{2}.$$

■

Exercise 1.4. Are there other choices for a, b, c, d for which the identity would hold?

Example 1.5. Let n be a positive integer. Show that

$$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \cdots + x + 1).$$

²and try to see what conditions get imposed on p, q . It may happen that the conditions that get imposed, may suggest that there are no such p, q . However, it may also happen that we would be able to find out which p, q would work!

³It should be noted that p, q were assumed to exist such that $p(2x + 3) + q(2x + 5) = 2x - 7$ holds. Under this hypothesis, we obtained $p = 6, q = -5$. At this point, we cannot immediately conclude that $6(2x + 3) + (-5)(2x + 5) = 2x - 7$ holds (unless we verify it), because if we do so, then we would do it under the same hypothesis.

- Even then, what would go wrong with that?
- Can a hypothesis (possibly combined with some of its consequences) be a justification for itself to hold? Think about this point.

Solution 5. Note that

$$\begin{aligned}
 & (x-1)(x^{n-1} + x^{n-2} + \cdots + x + 1) \\
 &= x(x^{n-1} + x^{n-2} + \cdots + x + 1) - (x^{n-1} + x^{n-2} + \cdots + x + 1) \\
 &= x^n + x^{n-1} + \cdots + x^2 + x - (x^{n-1} + x^{n-2} + \cdots + x + 1) \\
 &= x^n - 1.
 \end{aligned}$$

■

The exercise below relies on Example 1.5.

Exercise 1.6 (Moscow Mathematical Olympiad 2015 Grade 9 P6). Do there exist two polynomials with integer coefficients such that each of them has a coefficient with absolute value exceeding 2015, but no coefficient of their product has absolute value exceeding 1?

Summary — Try to come up with enough polynomials $g_1(x), g_2(x), g_3(x), \dots$ and $h_1(x), h_2(x), h_3(x), \dots$ such that each of the products $g_1 g_2 g_3 \dots$ and $h_1 h_2 h_3 \dots$ have at least one coefficient which is **large in absolute value**, and all the coefficients of the product $(g_1 g_2 g_3 \dots)(h_1 h_2 h_3 \dots)$ are at most 1 in absolute value.

Walkthrough —

- (a) Try to come up with a polynomial $P(x)$ whose coefficients are at most 1 in absolute value, and it can be written as a product of enough factors (say $f_1(x), f_2(x), \dots$) such that each of such factor $f_i(x)$ admits a decomposition into the product of two polynomials $g_i(x)$ and $h_i(x)$.
- (b) Can you make sure that the product of the g_i 's, and the product of the h_i 's have to have at least one large coefficient?
- (c) For instance, would taking $g_1(x) = g_2(x) = g_3(x) = \cdots = 1 - x$ work for some suitable choice of $h_1(x), h_2(x), \dots$?
- (d) Does taking

$$\begin{aligned}
 h_1(x) &= 1 + x, \\
 h_2(x) &= 1 + x + x^2, \\
 h_3(x) &= 1 + x + x^2 + x^3,
 \end{aligned}$$

etc. work?

- (e) Note that the product of enough g_i 's would have a large coefficient (namely, the coefficient of the second largest power of x). On the other hand, the product of enough h_i 's would have a large coefficient (namely, the coefficient of the power of x).

(f) What can be said about the absolute value of the coefficients of the product of these two products?

The above seems to work except that having a control on the coefficients of the product $(g_1g_2g_3\ldots)(h_1h_2h_3\ldots)$ seems hard⁴.

Solution 6. Consider the polynomial

$$P(x) = (1-x)(1-x^2)(1-x^4)(1-x^8)\cdots(1-x^{2^{2016}}).$$

Since

$$1 + 2 + 2^2 + 2^3 + \cdots + 2^{n-1} < 2^n,$$

it follows that the coefficients of $P(x)$ are at most 1 in absolute value. Note that

$$P(x) = Q(x)R(x)$$

holds where

$$Q(x) = (1-x)^{2017},$$

$$R(x) = (1+x)(1+x+x^2+x^3)\cdots(1+x+x^2+\cdots+x^{2^{2016}-1}).$$

The coefficient of x^{2016} in $Q(x)$ is equal to 2017, and the coefficient of x in $R(x)$ is equal to 2016. This completes the proof. ■

Exercise 1.7 (All-Russian Mathematical Olympiad 2007 Grade 10 Day 1 P2, AoPS, by A. Khrabrov). Consider a polynomial $P(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n$. Let m denote the minimum of the integers

$$a_0, a_0 + a_1, \dots, a_0 + a_1 + \cdots + a_n.$$

Prove that $P(x) \geq mx^n$ for all $x \geq 1$.

Walkthrough —

(a)

Solution 7. Note that

$$\begin{aligned} P(x) &= a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n \\ &= a_0(x^n - x^{n-1}) + (a_0 + a_1)(x^{n-1} - x^{n-2}) \\ &\quad + \cdots + (a_0 + a_1 + \cdots + a_{n-1})(x - 1) + (a_0 + a_1 + \cdots + a_n) \end{aligned}$$

holds, which implies that

$$P(x) - mx^n$$

⁴Is it because it fails?

$$\begin{aligned}
&= a_0(x^n - x^{n-1}) + (a_0 + a_1)(x^{n-1} - x^{n-2}) \\
&\quad + \cdots + (a_0 + a_1 + \cdots + a_{n-1})(x - 1) + (a_0 + a_1 + \cdots + a_n) \\
&\quad - m(x^n - x^{n-1}) - m(x^{n-1} - x^{n-2}) - \cdots - m(x - 1) - m \\
&= (a_0 - m)(x^n - x^{n-1}) + (a_0 + a_1 - m)(x^{n-1} - x^{n-2}) \\
&\quad + \cdots + (a_0 + a_1 + \cdots + a_{n-1} - m)(x - 1) + (a_0 + a_1 + \cdots + a_n - m).
\end{aligned}$$

It follows that for any $x \geq 1$,

$$P(x) \geq mx^n.$$

■

§1.2 Division algorithm

Example 1.8. Prove that the polynomial $x^{44} + x^{33} + x^{22} + x^{11} + 1$ is divisible by the polynomial $x^4 + x^3 + x^2 + x + 1$.

Solution 8. Note that

$$\begin{aligned}
&x^{44} + x^{33} + x^{22} + x^{11} + 1 \\
&= x^{40} \cdot x^4 + x^{30} \cdot x^3 + x^{20} \cdot x^2 + x^{10} \cdot x + 1 \\
&= (x^{40} - 1)x^4 + (x^{30} - 1)x^3 + (x^{20} - 1)x^2 + (x^{10} - 1)x \\
&\quad + x^4 + x^3 + x^2 + x + 1.
\end{aligned}$$

Hence, to prove that the polynomial $x^{44} + x^{33} + x^{22} + x^{11} + 1$ is divisible by $x^4 + x^3 + x^2 + x + 1$, it suffices to show that $x^4 + x^3 + x^2 + x + 1$ divides the polynomials

$$x^{40} - 1, x^{30} - 1, x^{20} - 1, x^{10} - 1.$$

Since

$$\begin{aligned}
x^5 - 1 &= (x - 1)(x^4 + x^3 + x^2 + x + 1), \\
x^{10} - 1 &= (x^5 - 1)(x^5 + 1),
\end{aligned}$$

it follows that $x^4 + x^3 + x^2 + x + 1$ divides $x^{10} - 1$. Moreover, the polynomial x^{10-1} divides all of

$$x^{40} - 1, x^{30} - 1, x^{20} - 1, x^{10} - 1.$$

Hence, $x^4 + x^3 + x^2 + x + 1$ divides the polynomials

$$x^{40} - 1, x^{30} - 1, x^{20} - 1, x^{10} - 1.$$

■

Exercise 1.9. Show that the polynomial $x^{580} + x^{390} + x^{326} + x^{262} + x^{198} + x^{134} + 1$ is divisible by $x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$.

Example 1.10. Determine the remainder obtained upon dividing x^{100} by $x^2 - 3x + 2$.

Solution 9. Let $q(x)$ (resp. $r(x)$) denote the quotient (resp. the remainder) obtained upon dividing x^{100} by $x^2 - 3x + 2$. Note that $r(x)$ is a linear polynomial, i.e. $r(x) = ax + b$ for some real numbers a, b . Then we have

$$x^{100} = q(x)(x^2 - 3x + 2) + r(x).$$

Substituting $x = 1$, it yields

$$1 = r(1) = a + b.$$

Similarly, substituting $x = 2$, it gives

$$2^{100} = r(2) = 2a + b.$$

This shows that

$$a = 2^{100} - 1, \quad b = 1 - a = 2 - 2^{100}.$$

Hence, the remainder obtained upon dividing x^{100} by $x^2 - 3x + 2$ is equal to

$$(2^{100} - 1)x + 2 - 2^{100}.$$

■

Example 1.11. Factorize the polynomials

$$x(y - z)^3 + y(z - x)^3 + z(x - y)^3, \quad xy(x^2 - y^2) + yz(y^2 - z^2) + zx(z^2 - x^2)$$

into products of linear polynomials.

§1.3 Bézout's theorem

Example 1.12. Let $f(x), g(x)$ be polynomials with integer coefficients and with no common factors. Show that $\gcd(f(n), g(n))$ attains only finitely many values as n varies over the integers.

§1.4 Even and odd polynomials

Exercise 1.13 (Moscow Mathematical Olympiad First Round 1946 Grades 7-8 P5). Prove that after completing the multiplication and collecting the terms

$$(1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100})$$

has no monomials of odd degree.

Summary — What happens if x is replaced by $-x$?

Walkthrough —

(a)

Solution 10. Let $P(x)$ denote the polynomial

$$(1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100}).$$

Note that $P(x) = P(-x)$. By the Claim below, it follows that $P(x)$ has no monomials of odd degree.

Claim — Let $Q(x)$ be a polynomial satisfying $Q(x) = Q(-x)$. Then $Q(x)$ has no monomials of odd degree.

Proof of the Claim. Note that

$$Q(x) = \frac{Q(x) + Q(-x)}{2} + \frac{Q(x) - Q(-x)}{2}$$

holds. Using $Q(x) = Q(-x)$, it follows that $Q(x) = \frac{Q(x) + Q(-x)}{2}$. Consequently, $Q(x)$ has no monomials of odd degree. $\square$

Remark. The above decomposition of $Q(x)$ is a special case of general phenomena^a.

^aCan you think of a few? Which **general phenomena** is referred to?!

Remark. The above solution is more elegant, and less cumbersome. Moreover, it also highlights the underlying reason, whereas the solution below obscures the conceptual viewpoint.

Solution 11. One can multiply the polynomials to note that

$$\begin{aligned} & 1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100} \\ &= 1 - x + x^2(1 - x) + x^4(1 - x) + x^6(1 - x) + \cdots + x^{98}(1 - x) + x^{100} \\ &= (1 - x)(1 + x^2 + x^4 + x^6 + \cdots + x^{98}) + x^{100}. \end{aligned}$$

Using this, we obtain

$$\begin{aligned} & (1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\ &= ((1 - x)(1 + x^2 + x^4 + x^6 + \cdots + x^{98}) + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\ &= (1 - x)(1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 + x + x^2 + \cdots + x^{99} + x^{100}) \end{aligned}$$

$$\begin{aligned}
& + x^{100}(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
& = (1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 - x^{101}) \\
& \quad + x^{100}(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
& = (1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 - x^{101}) \\
& \quad + x^{100}(x + x^3 + x^5 + \cdots + x^{99}) \\
& \quad + x^{100}(1 + x^2 + x^4 + \cdots + x^{98} + x^{100}) \\
& = (1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 - x^{101}) \\
& \quad + x^{101}(1 + x^2 + x^4 + x^6 + \cdots + x^{98}) \\
& \quad + x^{100}(1 + x^2 + x^4 + \cdots + x^{98} + x^{100}) \\
& = 1 + x^2 + x^4 + x^6 + \cdots + x^{98} + x^{100}(1 + x^2 + x^4 + \cdots + x^{98} + x^{100}),
\end{aligned}$$

which has no monomial of odd degree. ■

The following exercise is quite similar to the Claim proved in the solution to ??.

Exercise 1.14. Let $Q(x)$ be a polynomial satisfying $Q(x) = -Q(-x)$. Then $Q(x)$ has no monomials of even degree.

Example 1.15. Let n be an even positive integer, and let $p(x)$ be a polynomial of degree n such that $p(k) = p(-k)$ for $k = 1, 2, \dots, n$. Prove that there is a polynomial $q(x)$ such that $p(x) = q(x^2)$.

Walkthrough —

- (a) Note that the polynomial $p(x) - p(-x)$ has degree $< n$ because n is even. Observe that it has at least n roots.

Remark. What would happen if n is not assumed to be even?

Solution 12. Note that the polynomial $p(x) - p(-x)$ has degree at most n , and it vanishes at the integers $-n, \dots, -2, 1, 0, 1, 2, \dots, n$. Thus, it has at least $2n + 1$ roots. It follows that the polynomial is identically zero, that is, the polynomials $p(x), p(-x)$ are equal. This implies that $p(x)$ is equal to $q(x^2)$ for some polynomial $q(x)$. ■

Exercise 1.16 (Tournament of Towns Spring 2014, Senior A Level P7, by D. A. Zvonkin). Consider a polynomial $P(x)$ such that

$$P(0) = 1, \quad (P(x))^2 = 1 + x + x^{100}Q(x),$$

where $Q(x)$ is also a polynomial. Prove that in the polynomial $(P(x) + 1)^{100}$, the coefficient of x^{99} is zero.

Walkthrough —

- (a) Since $P(x)^2$ is congruent to $1+x$ modulo x^{100} , show that $(P(x)+1)^{100} + (1-P(x))^{100}$ is congruent to a polynomial of degree 50 in $1+x$ modulo x^{100} .
- (b) Prove that $(P(x)+1)^{100}$ is congruent to a polynomial of degree 50 in $1+x$ modulo x^{100} .

Solution 13. Note that

$$(P(x)+1)^{100} + (1-P(x))^{100}$$

is a polynomial in $P(x)^2$ of degree 50. Given three polynomials $f(x), g(x), h(x)$ having complex coefficients, with $h(x) \neq 0$, we say that **$f(x)$ is congruent to $g(x)$ modulo $h(x)$** if $h(x)$ divides $f(x) - g(x)$, that is, $f(x) - g(x)$ is the product of $h(x)$ and a polynomial in x with complex coefficients. Since $P(x)^2$ is congruent to $1+x$ modulo x^{100} , it follows that $(P(x)+1)^{100} + (1-P(x))^{100}$ is congruent to a polynomial of degree 50 in $1+x$ modulo x^{100} . Using that $P(x) \equiv 1 \pmod{x}$, we obtain that $(P(x)+1)^{100}$ is congruent to a polynomial of degree 50 in $1+x$ modulo x^{100} . This shows that the coefficient of x^{99} in $(P(x)+1)^{100}$ is zero. ■

§1.5 Factorization and roots

Example 1.17. Let a, b, c be three distinct real numbers. Show that

$$\frac{(a-x)(b-x)}{(a-c)(b-c)} + \frac{(b-x)(c-x)}{(b-a)(c-a)} + \frac{(c-x)(a-x)}{(c-b)(a-b)} = 1.$$

Walkthrough — Can a polynomial having degree at most two admit more than two distinct roots?

Exercise 1.18 (All-Russian Mathematical Olympiad 2011 Grade 9 Day 1 P1, AoPS, by A. Khrabrov). Let $P(x)$ be a monic quadratic polynomial such that $P(x)$ and $P(P(P(x)))$ admit a common root. Prove that $P(0) \cdot P(1) = 0$.

Walkthrough —

- (a) Show that $P(P(0))$ is equal to 0.
- (b) Show that $P(P(0)) = P(0) \cdot P(1)$.

Solution 14. Let α denote a common root of the polynomials $P(x), P(P(P(x)))$. Using

$$P(P(P(\alpha))) = P(\alpha) = 0,$$

it follows that

$$P(P(0)) = 0.$$

Since $P(x)$ is a monic quadratic polynomial, we obtain that

$$P(P(0)) = P(0) \cdot P(1).$$

This shows that $P(0) \cdot P(1) = 0$. ■

Exercise 1.19 (USAMO 1975 P3, AoPS). A polynomial $P(x)$ of degree n satisfies

$$P(k) = \frac{k}{k+1} \quad \text{for } k = 0, 1, 2, \dots, n.$$

Find $P(n+1)$.

Walkthrough —

(a) Consider the polynomial $(x+1)P(x) - x$.

Solution 15. Note that $xP(x+1) - x$ is a polynomial of degree $n+1$, and it vanishes at the $n+1$ integers $0, 1, 2, \dots, n$. It follows that

$$(x+1)P(x) - x = cx(x-1)(x-2)\dots(x-n)$$

for some nonzero real number c . Substituting $x = -1$ yields

$$-1 = (-1)^{n+1}c(n+1)!,$$

which gives $c = \frac{(-1)^n}{(n+1)!}$. This implies that

$$(n+2)P(n+1) = n+1 + (-1)^n,$$

and consequently,

$$P(n+1) = \frac{n+1 + (-1)^n}{n+2}.$$
■

Exercise 1.20 (International Mathematics Competition 2008 Day 2 P4). Let $f(x), g(x)$ be nonconstant polynomials with integer coefficients such that $g(x)$ divides $f(x)$. Prove that if the polynomial $f(x) - 2008$ has at least 81 distinct integer roots, then the degree of $g(x)$ is greater than 5.

Walkthrough —

(a)

Solution 16. Without loss of generality, we may assume that $g(x)$ has content 1, that is, the greatest common divisor of its coefficients is 1. Since $g(x)$ divides $f(x)$, there exists a polynomial $h(x)$ with rational coefficients such that $f(x) = g(x)h(x)$. Since $g(x)$ is a primitive polynomial with integer coefficients, by Gauss's lemma, the polynomial $h(x)$ also has integer coefficients. Denote the set of integer roots of $f(x) - 2008$ by S . Note that for each $a \in S$, we have $f(a) = 2008$, and thus $g(a)$ divides 2008. Since $2008 = 2^3 \cdot 251$, the number of positive integer divisors of 2008 is 8. Therefore, the polynomial $g(x)$ can take at most 16 distinct integer values on the set S . By hypothesis, the set S has at least 81 elements. By the pigeonhole principle, there exists a divisor d of 2008 such that the equation $g(x) = d$ has at least $\lceil 81/16 \rceil = 6$ distinct integer solutions. This shows that the polynomial $g(x) - d$ has at least 6 distinct integer roots, and hence its degree is at least 6. This implies that the degree of $g(x)$ is greater than 5, as required. ■

Exercise 1.21 (Asian Pacific Mathematics Olympiad 2009 P2, AoPS). Let a_1, a_2, a_3, a_4, a_5 be real numbers satisfying the following equations:

$$\frac{a_1}{k^2 + 1} + \frac{a_2}{k^2 + 2} + \frac{a_3}{k^2 + 3} + \frac{a_4}{k^2 + 4} + \frac{a_5}{k^2 + 5} = \frac{1}{k^2}$$

for $k = 1, 2, 3, 4, 5$. Find the value of

$$\frac{a_1}{37} + \frac{a_2}{38} + \frac{a_3}{39} + \frac{a_4}{40} + \frac{a_5}{41}.$$

(Express the value in a single fraction.)

Solution 17. Consider the polynomial

$$\begin{aligned} P(x) = & a_1x(x+2)(x+3)(x+4)(x+5) \\ & + a_2x(x+1)(x+3)(x+4)(x+5) \\ & + a_3x(x+1)(x+2)(x+4)(x+5) \\ & + a_4x(x+1)(x+2)(x+3)(x+5) \\ & + a_5x(x+1)(x+2)(x+3)(x+4) \\ & - (x+1)(x+2)(x+3)(x+4)(x+5). \end{aligned}$$

Note that $P(x)$ is a polynomial of degree at most 5, and it vanishes at $1^2, 2^2, 3^2, 4^2, 5^2$. Thus, we can write

$$P(x) = c(x-1)(x-4)(x-9)(x-16)(x-25)$$

for some real number c . Note that

$$P(0) = -120.$$

On the other hand,

$$\begin{aligned} P(0) &= c(-1)(-4)(-9)(-16)(-25) \\ &= -14400c. \end{aligned}$$

Equating the two expressions for $P(0)$, we get

$$c = \frac{1}{120}.$$

Note that

$$\begin{aligned} S &= \frac{a_1}{37} + \frac{a_2}{38} + \frac{a_3}{39} + \frac{a_4}{40} + \frac{a_5}{41} \\ &= \frac{P(36)}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} + \frac{1}{36} \\ &= \frac{1}{120} \times \frac{(36-1)(36-4)(36-9)(36-16)(36-25)}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} + \frac{1}{36} \\ &= \frac{1}{120} \times \frac{35 \cdot 32 \cdot 27 \cdot 20 \cdot 11}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} + \frac{1}{36} \\ &= \frac{55440}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} + \frac{37 \cdot 38 \cdot 39 \cdot 40 \cdot 41}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} \\ &= \frac{55440 + 89927760}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} \\ &= \frac{89983200}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} \\ &= \frac{187465}{6744582}. \end{aligned}$$

■

Example 1.22. Let $P(x)$ be a polynomial of degree $\leq n$ having rational coefficients. Suppose $P(k) = \frac{1}{k}$ holds for $1 \leq k \leq n+1$. Determine $P(0)$.

Solution 18. Note that $xP(x) - 1$ is a polynomial of degree at most $n+1$, and it vanishes at $1, 2, \dots, n+1$. This implies that

$$xP(x) - 1 = c(x-1)(x-2)\dots(x-n-1)$$

holds for some rational number c . Substituting $x = 0$, we obtain

$$c = \frac{(-1)^n}{(n+1)!}.$$

Differentiating and dividing by $xP(x) - 1$, we obtain

$$\frac{xP'(x) + P(x)}{xP(x) - 1} = \frac{1}{x-1} + \frac{1}{x-2} + \dots + \frac{1}{x-n-1},$$

which yields

$$P(0) = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

■

Example 1.23. Let $g(x)$ and $h(x)$ be polynomials with real coefficients such that

$$g(x)(x^2 - 3x + 2) = h(x)(x^2 + 3x + 2)$$

and $f(x) = g(x)h(x) + (x^4 - 5x^2 + 4)$. Prove that $f(x)$ has at least four real roots.

Solution 19. Note that $g(x)$ and $h(x)$ satisfy

$$g(x)(x-1)(x-2) = h(x)(x+1)(x+2),$$

which shows that

$$g(-1), g(-2), h(1), h(2)$$

are equal to 0. Also note that

$$\begin{aligned} x^4 - 5x^2 + 4 &= (x^2 - 1)(x^2 - 4) \\ &= (x-1)(x+1)(x+2)(x-2). \end{aligned}$$

Hence, the polynomials $g(x)h(x)$ and $x^4 - 5x^2 + 4$ vanish at $1, -1, 2, -2$. Consequently, f also vanishes at these four points. ■

Example 1.24. Let $P(x)$ be a polynomial with real coefficients such that $P(\sin \alpha) = P(\cos \alpha)$ for all $\alpha \in \mathbb{R}$. Show that $P(x) = Q(x^2 - x^4)$ for some polynomial $Q(x)$ with real coefficients.

Walkthrough —

- (a) Show that $P(x) = P(-x)$ for any $-1 \leq x \leq 1$, and hence $P(x) = f(x^2)$.
- (b) Deduce that $f(x) = f(1-x)$ for any $0 \leq x \leq 1$.
- (c) Using induction or otherwise, prove that $f(x) = g(x - x^2)$ for some polynomial $g(x)$ with real coefficients.

Example 1.25 (cf. [Moscow MO 1953 Grade 10](#), [India BMath 2006 P3](#)). [\[AE11, §1.7\]](#) Let n be a positive integer. Find the roots of the polynomial

$$P_n(X) = 1 + \frac{X}{1!} + \frac{X(X+1)}{2!} + \cdots + \frac{X(X+1) \cdots (X+n-1)}{n!}.$$

Walkthrough — Note that $P_1(X) = X + 1$ has -1 as its root, $P_2(X) = \frac{1}{2}(X+1)(X+2)$ has $-1, -2$ as its roots. Check that

$$P_3(X) = \frac{1}{3!}(X+1)(X+2)(X+3).$$

What happens for general n ?

Solution 20. We claim that the roots of $P_n(X)$ are $-1, -2, \dots, -n$ for any integer $n \geq 1$. Note that the claim holds for $n = 1$. Suppose the claim holds for some integer n . Comparing leading coefficients, it follows that

$$P_n(X) = \frac{1}{n!}(X+1)(X+2)\cdots(X+n).$$

Observe that

$$\begin{aligned} P_{n+1}(X) &= P_n(X) + \frac{X(X+1)\cdots(X+n)}{(n+1)!} \\ &= \frac{1}{n!}(X+1)(X+2)\cdots(X+n) + \frac{X(X+1)\cdots(X+n)}{(n+1)!} \\ &= \frac{1}{(n+1)!}(X+1)(X+2)\cdots(X+n)(n+1+X). \end{aligned}$$

Hence the roots of P_{n+1} are $-1, -2, \dots, -(n+1)$. The claim follows by induction. ■

Example 1.26. Show that any odd degree polynomial with real coefficients has at least one real root.

Exercise 1.27 (Putnam 1999 A2, AoPS). Show that for some fixed positive integer n , we can always express a polynomial with real coefficients which is nowhere negative as a sum of the squares of n polynomials.

Walkthrough —

- (a) Show that the real roots of P have even multiplicity.
- (b) Conclude that P can be expressed as a product of monic quadratic polynomials with real coefficients having nonreal roots, and even powers of linear polynomials with real coefficients.
- (c) Show that a monic quadratic polynomial with real coefficients having nonreal roots is the sum of the squares of two polynomials with real coefficients.

Solution 21. Note that if P is a constant polynomial, then it is clear. Henceforth, let us assume that P is a nonconstant polynomial.

Claim — The polynomial P can be written as the product of polynomials, each of which can be expressed as the sum of the squares of two polynomials with real coefficients.

Proof of the Claim. Since P has real coefficients, it follows that if $\alpha \in \mathbb{C} \setminus \mathbb{R}$ is a root of P , then so is $\bar{\alpha}$. Thus, the nonreal complex roots of P form pairs of complex conjugates. Note that

$$(x - \alpha)(x - \bar{\alpha}) = (x - \operatorname{Re}(\alpha))^2 + \operatorname{Im}(\alpha)^2.$$

Decomposing P over the pairs of nonreal complex conjugate roots, and the real roots, it follows that P can be expressed as the product

$$cf(x) \prod_{a \in A} (x - a)^{m_a},$$

where c denote the leading coefficient of P , $f(x)$ denotes the product of (possibly no) quadratic polynomials of the form $(x - a)^2 + b^2$ with $a \in \mathbb{R}, b \in \mathbb{R} \setminus \{0\}$, and A denotes the set of real roots of P , and for an element $a \in A$, the multiplicity of a is denoted by m_a .

Evaluating P at a suitable real number (for instance, at $1 + \sum_{a \in A} a$ (resp. 1) if A is nonempty (resp. empty)), it follows that $c > 0$.

Let a be an element of A . Since A is finite, there exists a real number $\varepsilon > 0$ such that the interval $(a - \varepsilon, a + \varepsilon)$ contains no real roots of P other than a . If m_a were odd, then the sign of $P(x)$ would not remain constant as x ranges over in $(a - \varepsilon, a + \varepsilon) \setminus \{a\}$. Hence, it follows that m_a is even.

Since $c > 0$ and m_a is even for any $a \in A$, the Claim follows. $\square$

Claim — Let $f_1(x), g_1(x), f_2(x), g_2(x)$ be polynomials with real coefficients. Then the following holds.

$$\begin{aligned} & (f_1(x)^2 + g_1(x)^2)(f_2(x)^2 + g_2(x)^2) \\ &= (f_1(x)f_2(x) - g_1(x)g_2(x))^2 + (f_1(x)g_2(x) - f_2(x)g_1(x))^2 \end{aligned}$$

Proof of the Claim. Note that

$$\begin{aligned} & (f_1(x)^2 + g_1(x)^2)(f_2(x)^2 + g_2(x)^2) \\ &= f_1(x)^2 f_2(x)^2 + g_1(x)^2 g_2(x)^2 - 2f_1(x)f_2(x)g_1(x)g_2(x) \\ & \quad + f_1(x)^2 g_2(x)^2 + f_2(x)^2 g_1(x)^2 + 2f_1(x)g_2(x)f_2(x)g_1(x) \\ &= (f_1(x)f_2(x) - g_1(x)g_2(x))^2 + (f_1(x)g_2(x) + f_2(x)g_1(x))^2. \end{aligned}$$

$\square$

Combining the above Claims, and using induction, the result follows. $\blacksquare$

Example 1.28 (India RMO 2017a P3). Let $P(x) = x^2 + \frac{x}{2} + b$ and $Q(x) = x^2 + cx + d$ be two polynomials with real coefficients such that $P(x)Q(x) = Q(P(x))$ for all real x . Find all real roots of $P(Q(x)) = 0$.

Solution 22. Let α be a root of $P(x)$ in $\mathbb{C}$. Substituting $x = \alpha$ in $P(x)Q(x) = Q(P(x))$, we obtain $Q(0) = 0$, which shows that $d = 0$. This gives

$$P(x)(x^2 + cx) = P(x)(P(x) + c),$$

which yields $P(x) + c = x^2 + cx$. It follows that $c = \frac{1}{2}, b = -\frac{1}{2}$.

Note that the roots of $P(x)$ are $-1, \frac{1}{2}$. Hence, any root β of $P(Q(x)) = 0$ satisfies $Q(\beta) = -1$ or $Q(\beta) = \frac{1}{2}$. Since the discriminant of $Q(x) + 1$ is negative, it follows that if β is a real root of $P(Q(x)) = 0$, then $Q(\beta) = \frac{1}{2}$, which gives $\beta = -1$ or $\beta = 1/2$.

Note that $P(\frac{1}{2}) = 0$ and $-1, \frac{1}{2}$ are the roots of $Q(x) = 1/2$. This shows that the real roots of $P(Q(x)) = 0$ are $-1, \frac{1}{2}$. ■

§2 Viète's relations

Example 2.1 (India RMO 2012e P2). cf. [GA17, Problem 141] Let $P(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$ be a polynomial of degree $n \geq 3$. Knowing that $a_{n-1} = -\binom{n}{1}$ and $a_{n-2} = \binom{n}{2}$, and that all the roots of P are real, find the remaining coefficients. Note that $\binom{n}{r} = \frac{n!}{(n-r)!r!}$.

Solution 23. Let $\alpha_1, \alpha_2, \dots, \alpha_n$ denote the roots of P . Note that

$$\begin{aligned} & \sum_{1 \leq i, j \leq n, i \neq j} (\alpha_i - \alpha_j)^2 \\ &= 2(n-1) \sum_{1 \leq i \leq n} \alpha_i^2 - 2 \sum_{1 \leq i, j \leq n, i \neq j} \alpha_i \alpha_j \\ &= 2(n-1) \sum_{1 \leq i \leq n} \alpha_i^2 - 4 \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j \\ &= 2(n-1) \left(\sum_{1 \leq i \leq n} \alpha_i \right)^2 - 4(n-1) \left(\sum_{1 \leq i < j \leq n} \alpha_i \alpha_j \right) - 4 \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j \\ &= 2n^2(n-1) - 4n \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j \\ &= 2n^2(n-1) - 4n \binom{n}{2} \\ &= 0. \end{aligned}$$

Since $\alpha_1, \dots, \alpha_n$ are real, it follows that they are all equal. Using $\alpha_1 + \dots + \alpha_n = n$, we get

$$\alpha_1 = \alpha_2 = \dots = \alpha_n = 1.$$

This implies that $a_i = (-1)^i \binom{n}{i}$ for any $0 \leq i \leq n-1$. ■

Solution 24. Let $\alpha_1, \alpha_2, \dots, \alpha_n$ denote the roots of $P(x)$. Note that

$$\begin{aligned} & (\alpha_1 - 1)^2 + (\alpha_2 - 1)^2 + \dots + (\alpha_n - 1)^2 \\ &= \alpha_1^2 + \alpha_2^2 + \dots + \alpha_n^2 + n - 2(\alpha_1 + \alpha_2 + \dots + \alpha_n) \\ &= (\alpha_1 + \alpha_2 + \dots + \alpha_n)^2 - 2 \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j + n - 2(\alpha_1 + \alpha_2 + \dots + \alpha_n) \\ &= n^2 - n(n-1) + n - 2n \\ &= 0. \end{aligned}$$

So the roots $\alpha_1, \alpha_2, \dots, \alpha_n$ are all equal to 1. This implies that $P(x) = (x-1)^n$, and hence a_i is equal to $(-1)^i \binom{n}{i}$ for any $0 \leq i \leq n$. ■

Exercise 2.2 (All-Russian Mathematical Olympiad 2007 Grade 11 Day 2 P6, AoPS, by N. Agakhanov, I. Bogdanov). Do there exist nonzero reals a, b, c such that, for any $n > 3$, there exists a polynomial $P_n(x) = x^n + \dots + ax^2 + bx + c$, which has exactly n (not necessarily distinct) integral roots?

Walkthrough —

(a)

Solution 25. Assume that there exist nonzero reals a, b, c satisfying the given condition. Let $n > 3$ be a positive integer, and let $P_n(x) = x^n + \dots + ax^2 + bx + c$ be a polynomial with exactly n integral roots, which we denote by $r_1, r_2, \dots, r_n$, counted with multiplicities. Since c is nonzero, none of the roots $r_1, r_2, \dots, r_n$ is equal to zero. It follows that

$$\frac{1}{r_1}, \frac{1}{r_2}, \dots, \frac{1}{r_n}$$

are the roots of the polynomial

$$Q_n(x) = x^n P\left(\frac{1}{x}\right) = cx^n + bx^{n-1} + ax^{n-2} + \dots + x + 1.$$

By Vieta's formulas, we obtain

$$\frac{1}{r_1} + \dots + \frac{1}{r_n} = -\frac{b}{c}, \quad \sum_{1 \leq i < j \leq n} \frac{1}{r_i r_j} = \frac{a}{c}.$$

This implies that

$$\left(\frac{1}{r_1} + \cdots + \frac{1}{r_n}\right)^2 - 2 \sum_{1 \leq i < j \leq n} \frac{1}{r_i r_j} = \frac{b^2 - 2ac}{c^2},$$

and hence

$$\frac{1}{r_1^2} + \cdots + \frac{1}{r_n^2} = \frac{b^2 - 2ac}{c^2}.$$

Also note that

$$r_1 r_2 \cdots r_n = (-1)^n c.$$

It follows that

$$b^2 - 2ac \geq \frac{r_1^2 r_2^2 \cdots r_n^2}{r_1^2} + \cdots + \frac{r_1^2 r_2^2 \cdots r_n^2}{r_n^2} \geq n.$$

This shows that $n \leq b^2 - 2ac$ for any integer $n > 3$, which is a contradiction.

Therefore, there do not exist nonzero reals a, b, c satisfying the given condition. ■

§3 On the growth of polynomials

Example 3.1 (India BStat-BMath 2012). Show that the polynomial $x^8 - x^7 + x^2 - x + 15$ has no real root.

Summary — In absolute value, a higher degree polynomial dominates a smaller degree polynomial at arguments which are large enough in absolute value.

Solution 26. Let α be a real number. Let us consider the following cases.

1. $\alpha \geq 1$,
2. $\alpha \leq 0$,
3. $0 \leq \alpha \leq 1$.

If $\alpha \geq 1$, then

$$\begin{aligned} & \alpha^8 - \alpha^7 + \alpha^2 - \alpha + 15 \\ &= \alpha^7(\alpha - 1) + \alpha(\alpha - 1) + 15 \\ &\geq 15. \end{aligned}$$

If $\alpha \leq 0$, then

$$\alpha^8 - \alpha^7 + \alpha^2 - \alpha + 15$$

$$\begin{aligned}
&= \alpha^8 + (-\alpha^7) + \alpha^2 + (-\alpha) + 15 \\
&\geq 15.
\end{aligned}$$

If $0 \leq \alpha \leq 1$, then

$$\begin{aligned}
&\alpha^8 - \alpha^7 + \alpha^2 - \alpha + 15 \\
&= \alpha^8 + (1 - \alpha^7) + \alpha^2 + (1 - \alpha) + 13 \\
&\geq 13.
\end{aligned}$$

It follows that the polynomial $x^8 - x^7 + x^2 - x + 15$ has no real root. ■

Example 3.2. Does there exist a polynomial $P(x)$ with rational coefficients such that $\sin x = P(x)$ for all $x \geq 100$?

Solution 27. Suppose there exists a polynomial $P(x)$ with rational coefficients such that $\sin x = P(x)$ for all $x \geq 100$. It follows that $P(x)$ has absolute value at most 1 for all $x \geq 100$.

Claim — Let $f(x)$ be a nonconstant polynomial with real coefficients. Then for any given $M > 0$, there exists a real number $x_0 > 0$ such that

$$|f(x)| > M$$

for all real number x satisfying $|x| > x_0$.

Proof of the Claim. Write

$$f(x) = a_d x^d + a_{d-1} x^{d-1} + \cdots + a_1 x + a_0,$$

where $a_d, a_{d-1}, \dots, a_0$ lie in $\mathbb{R}$ and d denotes the degree of $f(x)$. Note that for any real α ,

$$\begin{aligned}
|f(\alpha)| &= |a_d \alpha^d + a_{d-1} \alpha^{d-1} + \cdots + a_1 \alpha + a_0| \\
&\geq |a_d \alpha^d| - |a_{d-1} \alpha^{d-1} + \cdots + a_1 \alpha + a_0| \\
&\geq |a_d \alpha^d| - |a_{d-1} \alpha^{d-1}| - \cdots - |a_1 \alpha| - |a_0| \\
&= \left(\frac{1}{d} |a_d \alpha^d| - |a_{d-1} \alpha^{d-1}| \right) \\
&\quad + \left(\frac{1}{d} |a_d \alpha^d| - |a_{d-2} \alpha^{d-2}| \right) \\
&\quad + \cdots + \left(\frac{1}{d} |a_d \alpha^d| - |a_1 \alpha| \right) \\
&\quad + \left(\frac{1}{d} |a_d \alpha^d| - |a_0| \right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{|a_d|}{d} |\alpha|^{d-1} \left(|\alpha| - \left| \frac{da_{d-1}}{a_d} \right| \right) \\
&+ \frac{|a_d|}{d} |\alpha|^{d-2} \left(|\alpha|^2 - \left| \frac{da_{d-2}}{a_d} \right| \right) \\
&+ \cdots + \frac{|a_d|}{d} |\alpha| \left(|\alpha|^{d-1} - \left| \frac{da_1}{a_d} \right| \right) \\
&+ \frac{|a_d|}{d} \left(|\alpha|^d - \left| \frac{da_0}{a_d} \right| \right).
\end{aligned}$$

Hence, for any given $M > 0$,

$$|f(x)| > M$$

holds for any real number x of large enough absolute value. Indeed, for any real number x satisfying

$$\begin{aligned}
|x| &> \left| \frac{da_{d-1}}{a_d} \right|, \\
|x|^2 &> \left| \frac{da_{d-2}}{a_d} \right|, \\
&\dots > \dots, \\
|x|^{d-1} &> \left| \frac{da_1}{a_d} \right|, \\
|x|^d &> \left| \frac{da_0}{a_d} \right| + \frac{Md}{|a_d|},
\end{aligned}$$

or equivalently, satisfying

$$x > \max \left\{ \left| \frac{da_{d-1}}{a_d} \right|, \left(\left| \frac{da_{d-2}}{a_d} \right| \right)^{1/2}, \dots, \left(\left| \frac{da_1}{a_d} \right| \right)^{1/(d-1)}, \left(\left| \frac{da_0}{a_d} \right| + \frac{Md}{|a_d|} \right)^{1/d} \right\},$$

the inequality $|f(x)| > M$ holds. The Claim follows by taking

$$x_0 = \max \left\{ \left| \frac{da_{d-1}}{a_d} \right|, \left(\left| \frac{da_{d-2}}{a_d} \right| \right)^{1/2}, \dots, \left(\left| \frac{da_1}{a_d} \right| \right)^{1/(d-1)}, \left(\left| \frac{da_0}{a_d} \right| + \frac{Md}{|a_d|} \right)^{1/d} \right\}.$$

□

By the above Claim, it follows that $P(x)$ is a constant polynomial. This shows that $\sin x$ is constant on the interval $[100, \infty)$, which is impossible since $\sin 100\pi \neq \sin 101\pi$ and $101\pi, 100\pi$ lie in $[100, \infty)$. This contradicts the assumption that there exists a polynomial $P(x)$ with rational coefficients such that $\sin x = P(x)$ for all $x \geq 100$.

Hence, there does not exist a polynomial $P(x)$ with rational coefficients such that $\sin x = P(x)$ for all $x \geq 100$. ■

Example 3.3. Show that a polynomial function from $\mathbb{R}$ to $\mathbb{R}$ can be expressed as the difference of two strictly increasing polynomial functions in infinitely many ways.

Example 3.4 (India RMO 2015c P3). Let $P(x)$ be a nonconstant polynomial whose coefficients are positive integers. If $P(n)$ divides $P(P(n) - 2015)$ for all natural numbers n , then prove that $P(-2015) = 0$.

Summary — In absolute value, a higher degree polynomial dominates a smaller degree polynomial at arguments which are large enough in absolute value.

Solution 28. Let $Q(x), R(x)$ be polynomials with rational coefficients such that

$$P(P(x) - 2015) = P(x)Q(x) + R(x)$$

and $R(x) = 0$ or $\deg R(x) < \deg P(x)$. Note that $P(n)$ is positive for all integer $n \geq 1$ since the coefficients of $P(x)$ are positive integers. By the given condition, it follows that $P(n)$ divides $R(n)$ for any integer $n \geq 1$.

Claim — Let $f(x), g(x)$ be two polynomials with real coefficients. Suppose $f(x)$ is a nonconstant polynomial with a positive leading coefficient, and $\deg g(x) < \deg f(x)$. Then there exists an integer $n_0 \geq 1$ such that

$$f(n) > g(n)$$

for any $n \geq n_0$.

Proof of the Claim. Note that it suffices to prove the Claim if $f(x)$ is a monomial, that is, a power of x . Indeed, write $f(x) = a_d x^d + a_{d-1} x^{d-1} + \dots + a_0$ with $a_0, \dots, a_d \in \mathbb{R}$ and d denoting the degree of f . Also write $g(x) = b_e x^e + b_{e-1} x^{e-1} + \dots + b_0$ with $b_0, \dots, b_e \in \mathbb{R}$ and e denoting the degree of g . Noting that $a_d > 0$, it follows that for a positive integer n , the inequality

$$a_d n^d + a_{d-1} n^{d-1} + \dots + a_0 > b_e n^e + b_{e-1} n^{e-1} + \dots + b_0$$

holds if

$$a_d n^d > \frac{a_{d-1}}{a_d} n^{d-1} + \dots + \frac{a_0}{a_d} + \frac{b_e}{a_d} n^e + \frac{b_{e-1}}{a_d} n^{e-1} + \dots + \frac{b_0}{a_d}$$

is satisfied, which can be concluded provided the Claim is known in the case when f is a monomial.

Let us assume that f is a monomial. Write $f(x) = x^d$ where d denotes the degree of f , and write $g(x) = b_e x^e + b_{e-1} x^{e-1} + \dots + b_0$ with $b_0, \dots, b_e \in \mathbb{R}$ and e denoting the degree of g . For any integer n , note that

$$f(n) - g(n) = \left(\frac{1}{e+1} n^d - b_e n^e \right)$$

$$\begin{aligned}
& + \left(\frac{1}{e+1} n^d - b_{e-1} n^{e-1} \right) \\
& + \cdots + \left(\frac{1}{e+1} n^d - b_0 \right) \\
& \geq \left(\frac{1}{e+1} n^d - |b_e| n^e \right) \\
& + \left(\frac{1}{e+1} n^d - |b_{e-1}| n^{e-1} \right) \\
& + \cdots + \left(\frac{1}{e+1} n^d - |b_0| \right).
\end{aligned}$$

Since $d \geq e$, it follows that there exists an integer $n_0 \geq 1$ such that

$$\frac{1}{e+1} n^d - |b_e| n^e, \frac{1}{e+1} n^d - |b_{e-1}| n^{e-1}, \dots, \frac{1}{e+1} n^d - |b_0|$$

are positive for any $n \geq n_0$. This proves the Claim. $\square$

By the above Claim, it follows that $R(x)$ is the zero polynomial. This implies that

$$P(P(x) - 2015) = P(x)Q(x).$$

Since $P(x)$ is a nonconstant polynomial, it has a root z in $\mathbb{C}$. Substituting $x = z$ yields

$$P(-2015) = 0.$$

■

Example 3.5. Let $f(x), g(x)$ be polynomials with integer coefficients. Assume that for infinitely many integers n , $g(n)$ is nonzero and divides $f(n)$. Show that $g(x)$ divides $f(x)$ in the ring of polynomials with rational coefficients.

Solution 29. By division algorithm, there exist polynomials $q(x), r(x)$ with rational coefficients such that

$$f(x) = g(x)q(x) + r(x),$$

where $r(x)$ is the zero polynomial or has degree less than that of $g(x)$. Since $q(x), r(x)$ have rational coefficients, there exists a positive integer m such that $mq(x), mr(x)$ have integer coefficients. Note that

$$mf(x) = g(x)(mq(x)) + mr(x).$$

For infinitely many integers n , we have $g(n) \neq 0$ and $g(n)$ divides $mr(n)$. Let us consider the case that $r(x)$ is not the zero polynomial. Since $r(x)$ has degree less than that of $g(x)$, there exists a positive integer N such that

$$|mr(n)| < |g(n)|$$

for all integers $n > N$. It follows that $mr(n) = 0$ for infinitely many integers n . Hence, $r(x)$ is the zero polynomial, which is a contradiction. Therefore, $r(x)$ is the zero polynomial. Since $f(x), g(x)$ have integer coefficients, and $g(x)$ divides $f(x)$ in the ring of polynomials with rational coefficients. ■

§4 Gauss's lemma

Exercise 4.1 (ELMO 2009, AoPS, proposed by Evan O'Dorney). Let a, b, c be positive integers such that $a^2 - bc$ is a square. Prove that $2a + b + c$ is not prime.

Walkthrough —

- (a) Consider the quadratic polynomial $p(x) = bx^2 + 2ax + c$.
- (b) Show that its discriminant is a perfect square.
- (c) Use Gauss's lemma to show that $p(x)$ can be factored into linear polynomials with integer coefficients.
- (d) Note that the roots of $p(x)$ are negative rationals.
- (e) Conclude that $p(x)$ can be factored into linear polynomials with positive integer coefficients.
- (f) Conclude that $p(1) = 2a + b + c$ is not a prime.

Solution 30. Consider the quadratic polynomial $p(x) = bx^2 + 2ax + c$ with integer coefficients. Since its discriminant is a perfect square, it follows that its roots are rational, that is, it can be factored over the rationals. By Gauss's lemma, $p(x)$ can be factored into linear polynomials with integer coefficients. Since the leading coefficient of $p(x)$ is positive, it follows that it can be factored into linear polynomials with integer coefficients and having positive leading coefficients. Note that the roots of $p(x)$ are negative rationals. This proves that $p(x)$ can be factored into linear polynomials with positive integer coefficients. Noting that $p(1) = 2a + b + c$, it follows that $2a + b + c$ is not a prime. ■

Remark. Note that in the above, one may prove that $p(x)$ can be factored into linear polynomials with integer coefficients without using Gauss's lemma, possibly by establishing the lemma in this specific case. In fact, the above problem could serve as an introduction to Gauss's lemma.

§5 Irreducibility

Example 5.1. [WH96, Problem 27] Let $p_1, \dots, p_n$ denote $n \geq 1$ distinct integers. Show that the polynomial

$$(x - p_1)^2(x - p_2)^2 \cdots (x - p_n)^2 + 1$$

cannot be expressed as the product of two non-constant polynomials with integral coefficients.

Solution 31. On the contrary, let us assume that the polynomial

$$P(x) := (x - p_1)^2(x - p_2)^2 \cdots (x - p_n)^2 + 1$$

can be expressed as the product of two non-constant polynomials $f(x), g(x)$ with integral coefficients.

Let us first establish the following Claims.

Claim — Replacing f, g by $-f, -g$ respectively (if necessary), we may assume that f, g take positive values at all real arguments.

Proof of the Claim. Note that the polynomial $P(x) - 1$ vanishes at $x = p_1, \dots, p_n$. Since the product of the leading coefficients of $f(x)$ and $g(x)$ is equal to the leading coefficient of $P(x)$, we may replace $f(x), g(x)$ by $-f(x), -g(x)$ respectively (if necessary) to assume that the leading coefficients of $f(x), g(x)$ are positive. Since $P = fg$ and P does not have a real root, it follows that the polynomials f, g do not have any real roots. At large enough real arguments, the polynomials f, g take positive values. Since f, g have no real roots, we conclude that they take positive values at all real arguments. $\square$

Claim — The polynomials f, g are of degree n . Moreover, these polynomials are equal.

Proof of the Claim. On the contrary, let us assume that the degrees of f, g are not equal. Interchanging f, g if necessary, we assume that $\deg(f) < \deg(g)$. Since the sum of the degrees of f, g is equal to $2n$, it follows that $\deg(f) < n$.

For any $1 \leq i \leq n$, the integers $f(p_i), g(p_i)$ are equal to 1 or -1 . Since f, g take positive values at all real arguments, we obtain $f(p_i) = 1$ for any $1 \leq i \leq n$. This shows that the polynomial $f - 1$ has at least n distinct roots. Using $\deg(f) < n$, we conclude that $f - 1$ is the zero polynomial, which is impossible since f is a non-constant polynomial. Therefore, the hypothesis that the degrees of f, g are not equal is not tenable. This completes the proof of the first part of the Claim.

Note that f, g are polynomials of degree n with equal leading coefficients. This shows that the polynomial $f(x) - g(x)$ has degree less than n and it vanishes at the n distinct points $p_1, \dots, p_n$. It follows that $f = g$. $\square$

Using the above Claim, note that

$$f(x)^2 - ((x - p_1)(x - p_2) \cdots (x - p_n))^2 = 1,$$

or equivalently,

$$(f(x) + (x - p_1)(x - p_2) \cdots (x - p_n))(f(x) - (x - p_1)(x - p_2) \cdots (x - p_n)) = 1,$$

which implies that the polynomials

$$f(x) + (x - p_1)(x - p_2) \cdots (x - p_n), f(x) - (x - p_1)(x - p_2) \cdots (x - p_n)$$

are constant polynomials, and both of them are equal. Consequently, the polynomial $(x - p_1)(x - p_2) \cdots (x - p_n)$ is the zero polynomial, which is impossible. This shows that the hypothesis that the given polynomial can be expressed as the product of two non-constant polynomials with integral coefficients is not tenable. This completes the proof. $\blacksquare$

Example 5.2. Let n be a positive integer. Show that the polynomial

$$(x - 1)(x - 2) \cdots (x - n) - 1$$

is irreducible over the field of rational numbers.

Solution 32. Note that the given polynomial is irreducible if $n = 1$. It suffices to consider the case $n \geq 2$. On the contrary, let us assume that the polynomial is reducible over the rationals. Then, by Gauss's lemma, it is also reducible over the integers. Thus, there exist non-constant polynomials $f(x), g(x)$ with integer coefficients such that

$$(x - 1)(x - 2) \cdots (x - n) - 1 = f(x)g(x).$$

Note that for any integer $1 \leq k \leq n$, we have

$$f(k)g(k) = -1.$$

This implies that both $f(k)$ and $g(k)$ are non-zero integers whose product is equal to -1 . Hence, for any integer $1 \leq k \leq n$, the pair $(f(k), g(k))$ is either $(-1, 1)$ or $(1, -1)$. This shows that the polynomial $f(x) + g(x)$, which has degree less than n , has at least n distinct roots. Hence, $f(x) + g(x)$ is the zero polynomial, which yields

$$(x - 1)(x - 2) \cdots (x - n) - 1 = f(x)g(x) = -f(x)^2.$$

This is a contradiction since the leading coefficient of the polynomial on the left-hand side is positive. This completes the proof. $\blacksquare$

Example 5.3. Let n be a positive integer with $n \neq 4$. Show that the polynomial

$$(x-1)(x-2)\cdots(x-n)+1$$

is irreducible over the field of rational numbers.

Solution 33. Note that the given polynomial is irreducible if $n = 1$. If $n = 2$, then the given polynomial is equal to

$$x^2 - 3x + 3,$$

which has no rational root, and hence is irreducible over the rationals. It suffices to consider the case $n \geq 3$ with $n \neq 4$. On the contrary, let us assume that the given polynomial is reducible over the rationals. Then, by Gauss's lemma, it is also reducible over the integers. Thus, there exist non-constant polynomials $f(x), g(x)$ with integer coefficients such that

$$(x-1)(x-2)\cdots(x-n)+1 = f(x)g(x).$$

Note that for any integer $1 \leq k \leq n$, we have

$$f(k)g(k) = 1.$$

This shows that the polynomial $f(x) - g(x)$, which has degree less than n , has at least n distinct roots. Hence, $f(x) - g(x)$ is the zero polynomial, which yields

$$(x-1)(x-2)\cdots(x-n)+1 = f(x)g(x) = f(x)^2.$$

This implies that n is an even positive integer. Since $n \neq 4$, it follows that $n \geq 6$. Note that

$$\begin{aligned} & f\left(n - \frac{1}{2}\right)^2 \\ &= \left(n - \frac{1}{2} - 1\right)\left(n - \frac{1}{2} - 2\right)\cdots\left(n - \frac{1}{2} - (n-1)\right)\left(n - \frac{1}{2} - n\right) + 1 \\ &= -\frac{1}{2}\left(1 - \frac{1}{2}\right)\left(2 - \frac{1}{2}\right)\cdots\left((n-1) - \frac{1}{2}\right) + 1 \\ &< 1 - \frac{1}{4}\left(2 - \frac{1}{2}\right)\cdots\left((n-1) - \frac{1}{2}\right) \\ &\leq 1 - \frac{1}{4}\left(2 - \frac{1}{2}\right)\left(3 - \frac{1}{2}\right)\left(4 - \frac{1}{2}\right) \quad (\text{since } n \geq 6) \\ &= 1 - \frac{3 \cdot 5 \cdot 7}{32} \\ &< 0, \end{aligned}$$

which is impossible. This shows that the given polynomial is irreducible over the rationals. ■

Remark. After obtaining

$$(x-1)(x-2)\cdots(x-n)+1=(f(x))^2,$$

one can also argue as follows to complete the proof. Note that n is an even positive integer. For $i \in \{1, -1\}$, let

$$S_i = \{k \in \{1, 2, \dots, n\} : f(k) = i\}.$$

Note that both S_1 and S_{-1} are disjoint subsets of $\{1, 2, \dots, n\}$ whose union is equal to $\{1, 2, \dots, n\}$. Moreover, each of the sets S_1 and S_{-1} contain at most $\frac{n}{2}$ elements since $f(x)$ is a non-constant polynomial. This shows that both S_1 and S_{-1} contain exactly $\frac{n}{2}$ elements.

Note that some element of one of the sets S_1, S_{-1} differs from some element of the other set at least by 3. Indeed, if S_i contains 1, then using that S_i contains $\frac{n}{2}$ elements and

$$1 + (n-3) > \frac{n}{2}$$

holds for $n \geq 6$, it follows that S_{-i} must contain an element greater than or equal to 4. This shows that there exist elements $a \in S_1$ and $b \in S_{-1}$ such that $|a-b| \geq 3$. This yields

$$f(a) - f(b) = 2$$

is divisible by $|a-b|$, which is impossible.

Remark. Note that

$$(x-1)(x-2)(x-3)(x-4)+1=(x^2-5x+4)(x^2-5x+6)+1=(x^2-5x+5)^2,$$

which is reducible over the rationals.

Example 5.4. Let p be an odd prime number. Show that the polynomial

$$P(x) = x^p - x + p$$

is irreducible over the field of rational numbers.

Solution 34. On the contrary, let us assume that the polynomial $P(x)$ is reducible over the rationals. Then, by Gauss's lemma, it is also reducible over the integers. Thus, there exist non-constant polynomials $f(x), g(x) \in \mathbb{Z}[x]$ such that

$$P(x) = f(x)g(x).$$

Since the leading coefficient of $P(x)$ is 1, by multiplying $f(x), g(x)$ by -1 if necessary, we may assume that both $f(x)$ and $g(x)$ are monic polynomials. Since p is a prime number, by reordering $f(x), g(x)$ if necessary, we may assume that $|f(0)| = p$. Let $\alpha_1, \dots, \alpha_k$ denote the roots of $f(x)$ over the

complex numbers (counted with multiplicities). Since f is a monic polynomial, by Viète's formulas, we have

$$|\alpha_1 \cdots \alpha_k| = |f(0)| = p.$$

In particular, at least one of the roots, say α , satisfies

$$|\alpha| \geq p^{1/k}.$$

Since $P(\alpha) = 0$, we have

$$\alpha^p - \alpha + p = 0,$$

which implies

$$p \geq |\alpha|^p - |\alpha| = |\alpha|(|\alpha|^{p-1} - 1) \geq p^{1/k}(p^{(p-1)/k} - 1) \geq p^{1/(p-1)}(p-1).$$

This shows that

$$p^{1/(p-1)} \leq 1 + \frac{1}{p-1},$$

which gives

$$p \leq \left(1 + \frac{1}{p-1}\right)^{p-1} < 3.$$

Since p is a prime number, we have $p = 2$. This is a contradiction since p is odd. This completes the proof. ■

Theorem 1 (Eisenstein's criterion)

Let

$$f(x) = a_n x^n + \cdots + a_1 x + a_0$$

be a polynomial with integer coefficients. Let p be a prime number and assume that

$$\begin{aligned} a_n &\not\equiv 0 \pmod{p}, \\ a_{n-1}, \dots, a_0 &\equiv 0 \pmod{p}, \\ a_0 &\not\equiv 0 \pmod{p^2} \end{aligned}$$

holds. Then $f(x)$ cannot be expressed as a product of two non-constant polynomials with rational coefficients.

Example 5.5. [Art91, Chapter 11, Exercise 4.10, p. 444] Let

$$f(x) = a_{2n+1}x^{2n+1} + a_{2n}x^{2n} + \cdots + a_1x + a_0$$

be a polynomial of degree $2n+1$ with integer coefficients. Let p be a prime number and assume that

$$a_{2n+1} \not\equiv 0 \pmod{p},$$

$$\begin{aligned} a_0, a_1, \dots, a_n &\equiv 0 \pmod{p^2}, \\ a_{n+1}, \dots, a_{2n} &\equiv 0 \pmod{p}, \\ a_0 &\not\equiv 0 \pmod{p^3}. \end{aligned}$$

Show that $f(x)$ cannot be expressed as a product of two non-constant polynomials with rational coefficients.

Example 5.6. For any prime p , show that there exist non-constant monic polynomials $f_p(x), g_p(x)$ with integer coefficients such that

$$x^4 - 10x^2 + 1 \equiv f_p(x)g_p(x) \pmod{p}$$

holds. Can the polynomial $x^4 - 10x^2 + 1$ be expressed as the product of two non-constant polynomials with rational coefficients?

Example 5.7. Prove that the polynomial $x^n + 4$ is irreducible over $\mathbb{Z}[x]$ if and only if n is not a multiple of 4.

Solution 35. If n is a multiple of 4, then we can write $n = 4k$ for some positive integer k . In this case, we have

$$x^n + 4 = x^{4k} + 4 = (x^{2k} - 2x^k + 2)(x^{2k} + 2x^k + 2),$$

which shows that $x^n + 4$ is reducible over $\mathbb{Z}[x]$.

Now, suppose that n is not a multiple of 4. We will show that $x^n + 4$ is irreducible over $\mathbb{Z}[x]$. Assume for the sake of contradiction that $x^n + 4$ is reducible over $\mathbb{Z}[x]$. Then we can write

$$x^n + 4 = f(x)g(x),$$

where $f(x), g(x) \in \mathbb{Z}[x]$ are non-constant polynomials with degrees less than n . Since the roots of $x^n + 4$ in $\mathbb{C}$ are of absolute value $4^{1/n}$, the roots of $f(x)$ are also of absolute value $4^{1/n}$. Let the degree of $f(x)$ be d . The absolute value of the constant term of $f(x)$ is then $(4^{1/n})^d = 2^{2d/n}$. Since the constant term of $f(x)$ is an integer, it follows that n divides $2d$. Since $d < n$, we must have that $n = 2d$. Since n is not a multiple of 4, it follows that d is odd. Thus, $f(x)$ is a monic polynomial of odd degree with integer coefficients. Hence, it has a real root, implying that $x^n + 4$ has a real root, which is impossible since n is even. This shows that $x^n + 4$ is irreducible over $\mathbb{Z}[x]$.

This completes the proof. ■

§6 Differentiation and multiple roots

Lemma 2

Let $P(x)$ be a polynomial with complex coefficients, and α be a complex number. If α is a root of $P(x)$ of multiplicity $k \geq 1$, then α is a root of $P'(x)$ of multiplicity $k - 1$. Moreover, α is a root of $P(x)$ having multiplicity at least $r \geq 2$ (i.e., $(x - \alpha)^r$ divides $P(x)$) if and only if it is a root of $P(x), P'(x), \dots, P^{(r-1)}(x)$, where $P^{(k)}(x)$ denotes the k -fold derivative of $P(x)$.

Solution 36. ■

Example 6.1 (Putnam 1956 B7, IMOSL 1981 Cuba). The polynomials $P(z)$ and $Q(z)$ with complex coefficients have the same set of numbers for their zeroes but possibly different multiplicities. The same is true of the polynomials $P(z) + 1$ and $Q(z) + 1$. **Assume that at least one of $P(z), Q(z)$ is nonconstant.** Prove that $P(z) = Q(z)$.

Walkthrough —

- (a) Assume that $\deg P \geq \deg Q$.
 (b) Denote these two set of roots by S_1, S_2 . Considering multiplicities, show that

$$2 \deg P - |S_1| - |S_2| \leq \deg P' = \deg P - 1,$$

which yields

$$|S_1| + |S_2| > \deg P.$$

- (c) Note that $P - Q$ vanishes at the elements of $S_1 \cup S_2$, which has size larger than the degree of $P - Q$.

Solution 37. On the contrary, let us assume that P, Q are not equal. Without loss of generality, let us assume that $\deg P \geq \deg Q$. Let n denote the degree of P . Let us denote by j the number of distinct roots of P , and by k the number of distinct roots of $P + 1$. Note that these two sets of roots are disjoint. Using the equality

$$P(z) - Q(z) = (P(z) + 1) - (Q(z) + 1),$$

it follows that $P - Q$ has at least $j + k$ distinct roots. Since $\deg P \geq \deg Q$, it follows that $\deg(P - Q) \leq \deg P = n$, and hence $j + k \leq n$.

Note that the derivative P' has at least $n - j$ roots common with P counted with multiplicity, and at least $n - k$ roots common with $P + 1$ counted with multiplicity. Since these two sets of roots are disjoint, it follows that

$$\deg P' \geq 2n - j - k \geq n.$$

Since $\deg P \geq \deg Q$ and one of P, Q is nonconstant, it follows that P is nonconstant, and hence $\deg P' = n - 1$. Thus, we have

$$n - 1 = \deg P' \geq n,$$

which is a contradiction. Thus, we obtain $P = Q$. ■

Theorem 3 (Gauss–Lucas)

Let P be a nonconstant polynomial with complex coefficients. Then, all the roots of P' lie in the convex hull of the set of roots of P .

Proof. Write P as

$$P(z) = a(z - z_1)(z - z_2) \cdots (z - z_n),$$

where a is a nonzero complex number, and $z_1, z_2, \dots, z_n$ are the roots of P counted with multiplicity. Note that

$$P'(z) = a \sum_{k=1}^n (z - z_1) \cdots (z - z_{k-1})(z - z_{k+1}) \cdots (z - z_n).$$

Thus, for any z which is a root of P' but not of P , we have

$$\begin{aligned} \frac{P'(z)}{P(z)} &= \sum_{k=1}^n \frac{1}{z - z_k} \\ &= \sum_{k=1}^n \frac{\overline{z - z_k}}{|z - z_k|^2}, \end{aligned}$$

which yields

$$\sum_{k=1}^n \frac{z}{|z - z_k|^2} = \sum_{k=1}^n \frac{z_k}{|z - z_k|^2}.$$

This shows that if z is a root of P' but not of P , then

$$z = \sum_{k=1}^n \frac{1}{\sum_{\ell=1}^n \frac{|z - z_k|^2}{|z - z_\ell|^2}} z_k,$$

which is a convex combination of the roots of P . Further, if z is a root of both P' and P , then z lies in the convex hull of the roots of P . This completes the proof. □

Exercise 6.2 (Putnam 2018 B2, AoPS). Let n be a positive integer, and let $f_n(z) = n + (n-1)z + (n-2)z^2 + \cdots + 2z^{n-2} + z^{n-1}$. Show that all the roots of $f_n(z)$ lie outside the closed unit disk $\{z \in \mathbb{C} : |z| \leq 1\}$.

Walkthrough —**(a)**

Solution 38. Let us assume that there exists a root z of f_n such that $|z| \leq 1$. Note that

$$(1 - z)f_n(z) = n - z - z^2 - \dots - z^{n-1} - z^n.$$

Using the triangle inequality, we have

$$\begin{aligned} 0 &= |(1 - z)f_n(z)| &= |n - z - z^2 - \dots - z^{n-1} - z^n| \\ &\geq n - |z + z^2 + \dots + z^{n-1} + z^n| \\ &\geq n - |z| - |z^2| - \dots - |z^{n-1}| - |z^n| \\ &> n - 1 - 1 - \dots - 1 - 1 \\ &= 0. \end{aligned}$$

This shows that $z, z^2, \dots, z^n$ have absolute value 1, and have the same argument, and hence $z = 1$. However, $f_n(1) = n + (n - 1) + (n - 2) + \dots + 2 + 1$, which is nonzero. This is a contradiction. Hence, all the roots of f_n lie outside the closed unit disk. ■

Solution 39. Let $g_n(z)$ denote the polynomial $z^n + z^{n-1} + \dots + z + 1$. Note that

$$g'_n(z) = nz^{n-1} + (n-1)z^{n-2} + \dots + 2z + 1 = z^{n-1}f_n\left(\frac{1}{z}\right).$$

Observe that the roots of the polynomials g'_n, f_n are nonzero, and a nonzero complex number z is a root of f_n if and only if $\frac{1}{z}$ is a root of g'_n . By the Gauss–Lucas theorem (Theorem 3), all the roots of g'_n lie in the convex hull of the set of roots of g_n . Note that the roots of g_n are precisely the n -th roots of unity other than 1. Since g_n has no multiple roots, it follows that g'_n does not vanish at any of the roots of g_n . Thus, all the roots of g'_n lie in the interior of the convex hull of the set of roots of g_n , and consequently, the roots of g'_n lie in the open unit disk. Therefore, all the roots of f_n lie outside the closed unit disk. ■

§7 Finite differences

Note that

$$k^2 - (k - 1)^2 = 2k - 1$$

holds for any integer k . In particular, given a positive integer n , we have

$$2^2 - 1^2 = 2 \cdot 2 - 1,$$

$$\begin{aligned}
3^2 - 2^2 &= 2 \cdot 3 - 1, \\
4^2 - 3^2 &= 2 \cdot 4 - 1, \\
&\dots = \dots, \\
n^2 - (n-1)^2 &= 2 \cdot n - 1.
\end{aligned}$$

Adding them, it follows that

$$n^2 - 1 = 2(2 + 3 + \dots + n) - (n - 1),$$

which yields

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}.$$

Consider the polynomial $P(x) = x^3$. Note that

$$P(k) - P(k-1) = 3k^2 - 3k + 1.$$

Using a similar argument as above, it follows that given a positive integer n , we have

$$\begin{aligned}
1^3 - 0^3 &= 3 \cdot 1^2 - 3 \cdot 1 + 1, \\
2^3 - 1^3 &= 3 \cdot 2^2 - 3 \cdot 2 + 1, \\
3^3 - 2^3 &= 3 \cdot 3^2 - 3 \cdot 3 + 1, \\
4^3 - 3^3 &= 3 \cdot 4^2 - 3 \cdot 4 + 1, \\
&\dots = \dots, \\
n^3 - (n-1)^3 &= 3 \cdot n^2 - 3 \cdot n + 1.
\end{aligned}$$

Adding them, it follows that

$$n^3 = 3(1 + 2^2 + 3^2 + \dots + n^2) - 3(1 + 2 + 3 + \dots + n) + n,$$

which yields

$$\begin{aligned}
1 + 2^2 + 3^2 + \dots + n^2 &= \frac{n^3 - n}{3} + (1 + 2 + 3 + \dots + n) \\
&= \frac{n^3 - n}{3} + \frac{n(n+1)}{2} \\
&= \frac{1}{6}n(n+1)(2n+1).
\end{aligned}$$

Example 7.1 (India RMO 2013b P3). Consider the expression

$$2013^2 + 2014^2 + 2015^2 + \dots + n^2.$$

Prove that there exists a natural number $n > 2013$ for which one can change a suitable number of plus signs to minus signs in the above expression to make the resulting expression equal 9999.

Summary — “Differentiating” a polynomial enough times makes it linear.

Walkthrough —

- (a) Consider the polynomial $P(k) = k^2$, and the polynomial $Q(k) := P(k) - (k - 1)$.
- (b) Since $Q(k)$ is a linear polynomial in k , the difference $R(k) := Q(k) - Q(k - 2)$ is a constant, that is, it does not depend on k .
- (c) Note that $R(k)$ is a **± 1 -linear combination**^a of four consecutive squares.
- (d) Does this help?

^aWhat does it mean?

Solution 40. Consider the polynomial $P(k) = k^2$, and the polynomial $Q(k) := P(k) - (k - 1)$. Since $Q(k)$ is a linear polynomial in k , the difference $R(k) := Q(k) - Q(k - 2)$ is a constant, that is, it does not depend on k . Indeed, $Q(k) = 2k - 1$, and $R(k) = 4$. Note that $R(k) = k^2 - (k - 1)^2 - (k - 2)^2 + (k - 3)^2$. Note that

$$2013^2 + 2014^2 + 2015^2 + 2016^2 + 2017^2 > 9999$$

holds, and the integers $2013^2 + 2014^2 + 2015^2 + 2016^2 + 2017^2$, 9999 are congruent modulo 4, that is, they differ by a multiple of 4. Let $m \geq 1$ be an integer such that

$$9999 = 2013^2 + 2014^2 + 2015^2 + 2016^2 + 2017^2 - 4m$$

holds. Since

$$-k^2 + (k + 1)^2 + (k + 2)^2 - (k + 3)^2 = -4,$$

it follows that

$$\begin{aligned} & 9999 \\ &= 2013^2 + 2014^2 + 2015^2 + 2016^2 + 2017^2 \\ &\quad - 2018^2 + 2019^2 + 2020^2 - 2021^2 \\ &\quad - \dots \\ &\quad - ((2018 + 4(m - 1))^2 + (2019 + 4(m - 1))^2 \\ &\quad \quad + (2020 + 4(m - 1))^2 - (2021 + 4(m - 1))^2). \end{aligned}$$

It follows that there exists a natural number $n = 2021 + 4(m - 1) > 2013$, for which one can change a suitable number of plus signs to minus signs in the expression

$$2013^2 + 2014^2 + 2015^2 + \dots + n^2$$

to make the resulting expression equal to 9999. ■

Exercise 7.2 (Bay Area MO 12 2016 P4). Find a positive integer N and $a_1, a_2, \dots, a_N$, where $a_k = 1$ or $a_k = -1$ for each $k = 1, 2, \dots, N$, such that

$$a_1 \cdot 1^3 + a_2 \cdot 2^3 + a_3 \cdot 3^3 + \dots + a_N \cdot N^3 = 20162016,$$

or show that this is impossible.

Summary — “Differentiating” a polynomial enough times makes it linear.

Walkthrough —

- (a) Consider the polynomial $P(k) := k^3$. Note that $R(k) := P(k) - P(k-1)$ is a quadratic polynomial in k .
- (b) Also note that $S(k) := R(k) - R(k-2)$ is a linear polynomial in k .

Solution 41. Consider the polynomial $P(k) := k^3$. Note that $R(k) := P(k) - P(k-1)$ is equal to $3k^2 - 3k + 1$. Also note that $S(k) := R(k) - R(k-2)$ is equal to $6(2k-2) - 6$. This gives $S(k) - S(k-4) = 48$. It follows that **some ± 1 -linear combination** of any given eight consecutive cubes is equal to 48. More specifically,

$$k^3 - (k-1)^3 - (k-2)^3 + (k-3)^3 - (k-4)^3 + (k-5)^3 + (k-6)^3 - (k-7)^3 = 48,$$

or equivalently,

$$-k^3 + (k+1)^3 + (k+2)^3 - (k+3)^3 - (k+4)^3 + (k+5)^3 + (k+6)^3 - (k+7)^3 = 48.$$

Note that 20162016 is divisible by 3 and 16. Since 3, 16 do not have any common prime factor, it follows that 20162016 is a multiple of 48. Write

$$f(k) = -k^3 + (k+1)^3 + (k+2)^3 - (k+3)^3 - (k+4)^3 + (k+5)^3 + (k+6)^3 - (k+7)^3.$$

Note that

$$f(1) + f(9) + f(17) + \dots + f(8m-7) = 20162016,$$

where m denotes the integer $20162016/48$. We conclude that one may take $N = 8m = 20162016/6 = 3360336$ so that the given condition holds. ■

Consider the finite difference operator Δ defined on the set of polynomials by

$$\Delta P(x) = P(x+1) - P(x).$$

For a polynomial $P(x)$, we define the r -th finite difference recursively as

$$\Delta^r P(x) = \Delta(\Delta^{r-1} P(x)),$$

with the convention that $\Delta^0 P(x) = P(x)$.

For an integer $k \geq 0$, define the binomial coefficient $\binom{x}{k}$ as

$$\binom{x}{k} = \frac{x(x-1)(x-2) \cdots (x-k+1)}{k!}$$

if $k \geq 1$, and $\binom{x}{0} = 1$. Note that $\binom{x}{k}$ is a polynomial in x of degree k .

Theorem 4

Let $P(x)$ be a polynomial of degree n with leading coefficient a . Then the n -th finite difference $\Delta^n P(x)$ is a constant polynomial and is equal to $a \cdot n!$, and the polynomial $P(x)$ can be expressed as

$$P(x) = \sum_{k=0}^n \Delta^k P(0) \binom{x}{k}.$$

Exercise 7.3 (Harvard-MIT Mathematics Tournament 2015 P1). The complex numbers x, y, z satisfy

$$\begin{aligned} xyz &= -4, \\ (x+1)(y+1)(z+1) &= 7, \\ (x+2)(y+2)(z+2) &= -3. \end{aligned}$$

Find, with proof, the value of $(x+3)(y+3)(z+3)$.

Walkthrough —

(a)

Solution 42. Let $P(t)$ denote the polynomial $(t+x)(t+y)(t+z)$. Note that

$$P(0) = -4, P(1) = 7, P(2) = -3.$$

This gives

$$\Delta P(0) = 11, \Delta P(1) = -10, \Delta^2 P(0) = -21, \Delta^3 P(0) = 6.$$

It follows that

$$\Delta^2 P(1) = \Delta^2 P(0) + \Delta^3 P(0) = -15,$$

which in turn gives

$$\Delta P(2) = \Delta P(1) + \Delta^2 P(1) = -25.$$

Finally, we have

$$P(3) = P(2) + \Delta P(2) = -28.$$



Example 7.4. Let n be an even positive integer, and P be a polynomial of degree n . Assume that $P(0) = 1$ and $P(j) = 2^{j-1}$ for all $j = 1, 2, \dots, n$. Prove that

$$P(n+2) = 2P(n+1) - 1.$$

Solution 43. Note that it suffices to prove the following claim.

Claim — Let n be an even positive integer, and P be a polynomial of degree n satisfying

$$P(0) = 1, P(j) = 2^{j-1}$$

for all $j = 1, 2, \dots, n$. Then the polynomial P satisfies

$$P(n+1) = 2^n, P(n+2) = 2^{n+1} - 1.$$

Proof. Let us first consider the case $n = 2$. Using

$$P(0) = 1, P(1) = 1, P(2) = 2,$$

we have

$$\Delta P(0) = 0, \Delta P(1) = 1,$$

and thus

$$\Delta^2 P(0) = 1.$$

Since P is a polynomial of degree 2, it follows that $\Delta^2 P(x)$ is a constant polynomial, and thus $\Delta^2 P(x) = 1$ for all integer x . Therefore, we have

$$\Delta P(2) = \Delta P(1) + \Delta^2 P(1) = 1 + 1 = 2,$$

and

$$\Delta P(3) = \Delta P(2) + \Delta^2 P(2) = 2 + 1 = 3.$$

This yields

$$P(3) = P(2) + \Delta P(2) = 2 + 2 = 4,$$

and

$$P(4) = P(3) + \Delta P(3) = 4 + 3 = 7.$$

This proves ⁵ the claim for $n = 2$.

Assume that the claim holds for some even positive integer k . Let us consider the case $n = k + 2$. Let P be a polynomial of degree $k + 2$ such that

$$P(0) = 1, P(j) = 2^{j-1}$$

⁵It is easier to **visualise** this using the following difference table:

x	0	1	2	3	4
$P(x)$	1	1	2	2 + 2	2 + 2 + 3
$\Delta P(x)$	0	1	2	3	
$\Delta^2 P(x)$	1	1	1		

for all $j = 1, 2, \dots, k + 2$. Note that ΔP takes the values

$$0, 1, 2, 4, \dots, 2^k$$

at $0, 1, 2, 3, \dots, k + 1$, respectively. This implies that $\Delta^2 P$ takes the values

$$1, 1, 2, 4, \dots, 2^{k-1}$$

at $0, 1, 2, 3, \dots, k$, respectively. Note that $\Delta^2 P$ is a polynomial of degree k . By the induction hypothesis, we have

$$\Delta^2 P(k + 1) = 2^k, \Delta^2 P(k + 2) = 2^{k+1} - 1.$$

This implies that

$$\Delta P(k + 2) = \Delta P(k + 1) + \Delta^2 P(k + 1) = 2^k + 2^k = 2^{k+1},$$

and

$$\Delta P(k + 3) = \Delta P(k + 2) + \Delta^2 P(k + 2) = 2^{k+1} + (2^{k+1} - 1) = 2^{k+2} - 1.$$

Therefore, we have

$$P(k + 3) = P(k + 2) + \Delta P(k + 2) = 2^{k+1} + 2^{k+1} = 2^{k+2},$$

and

$$P(k + 4) = P(k + 3) + \Delta P(k + 3) = (2^{k+2}) + (2^{k+2} - 1) = 2^{k+3} - 1.$$

This proves the claim for $n = k + 2$. By induction, the claim holds for all even positive integers n . □

■

§8 Size of the roots

Some of the following problems are from [Goy21].

Example 8.1. Let $f(x)$ and $g(x)$ be nonconstant polynomials with real coefficients such that $f(x^2 + x + 1) = f(x)g(x)$. Show that $f(x)$ has even degree.

Walkthrough — If the polynomial $f(x)$ admits a real root α , then note that $\alpha^2 + \alpha + 1$ is also a real root of $f(x)$ and $\alpha^2 + \alpha + 1 > \alpha$.

Solution 44. If the polynomial $f(x)$ admits a real root α , then note that $\alpha^2 + \alpha + 1$ is also a real root of $f(x)$ and $\alpha^2 + \alpha + 1 > \alpha$. Hence, if $f(x)$ possesses a real root, it would have real roots of arbitrarily large absolute value. Since $f(x)$ is nonconstant, it follows that $f(x)$ has no real roots. Since $f(x)$ is a polynomial with real coefficients, it follows that $f(x)$ is of even degree. ■

Exercise 8.2 (Moldova National Olympiad 2001 Grade 12 Day 1 P3, AoPS).

Find all polynomials $P(x)$ with real coefficients such that

$$P(x^2) = P(x)P(x-1)$$

for all $x \in \mathbb{R}$.

Walkthrough —

(a)

Solution 45. Let $P(x)$ be a polynomial with real coefficients satisfying the given functional equation. Note that the polynomials $P(x^2)$, $P(x)P(x-1)$ are equal.

Claim — Any root of $P(x)$ is equal to 0 or is a root of unity.

Proof of the Claim. Let α be a root of $P(x)$ in $\mathbb{C}$. It follows that α^2 is a root of $P(x)$. Since $P(x)$ has finitely many roots, and it vanishes at α^{2^n} for any positive integer n , we conclude that $\alpha = 0$ or α is a root of unity. □

Let β be a root of $P(x)$ in $\mathbb{C}$. Note that $(\beta+1)^2$ is a root of $P(x)$. This shows that if $P(x)$ has real root, then it has a nonnegative real root. Further, if $P(x)$ admits a nonnegative real root, then it admits infinitely many nonnegative real roots, which is impossible. This implies that $P(x)$ admits no real roots.

By the above claim, it follows that $(\beta+1)^2$ is equal to 0 or is a root of unity. Since $P(x)$ has no real roots, we obtain that $\beta+1$ is a root of unity. If $\beta+1=0$, then $\beta=-1$, which is impossible since $P(x)$ has no real roots. Otherwise, $\beta+1$ is a root of unity, which gives $(\beta+1)(\bar{\beta}+1)=1$, implying that $\beta+\bar{\beta}=-1$. Hence, β and $\bar{\beta}$ are the primitive third roots of unity.

Since $P(x)$ has real coefficients, it follows that $\bar{\beta}$ is also a root of $P(x)$. Since $P(x)$ has real coefficients, it follows that the multiplicity of the roots β and $\bar{\beta}$ of $P(x)$ are equal. Using that $P(x)$ has no real roots, we conclude that

$$P(x) = c(x^2 + x + 1)^k$$

for some $c \in \mathbb{R}$ and nonnegative integer k .

Using that $P(x^2)$ and $P(x)P(x-1)$ are equal polynomials, we obtain that $c^2 = c$, implying that $c = 0$ or $c = 1$.

Note that

$(x^2+x+1)^k((x-1)^2+(x-1)+1)^k = (x^2+x+1)^k(x^2-x+1)^k = ((x^2)^2+x^2+1)^k$ holds.

Therefore, the zero polynomial and the polynomials of the form $(x^2+x+1)^k$, where k is a nonnegative integer, are precisely all the polynomials satisfying the given condition. ■

Example 8.3. Find all polynomials $P(x)$ (with complex coefficients) satisfying

$$P(x)P(x+2) = P(x^2).$$

Summary — Note that if α is a root of P , then so are α^2 and $(\alpha-2)^2$. Considering absolute values, show that P cannot have a root other than 1. Conclude that $P(x) = c(x-1)^n$.

Walkthrough —

(a)

Solution 46. Let $P(x)$ be a polynomial with complex coefficients satisfying the given condition.

Claim — If P has a root $\alpha \neq 1$, then P has a root which has absolute value larger than the absolute value of α .

Proof of the Claim. Note that if $|\alpha| \leq 1$, then

$$\begin{aligned} |(\alpha-2)^2| &\geq (2-|\alpha|)^2 \\ &\geq 1 \\ &\geq |\alpha|. \end{aligned}$$

Moreover, if $|(\alpha-2)^2| = |\alpha|$ holds, then $|\alpha| = 1$ and $\alpha = 2r$ for some real number $r \geq 0$, and hence, $\alpha = 1$. This shows that if $\alpha \neq 1$, and $|\alpha| \leq 1$, then $|(\alpha-2)^2| > |\alpha|$ holds. Also note that if $|\alpha| > 1$, then $|\alpha^2| > |\alpha|$ holds. Since $\alpha^2, (\alpha-2)^2$ are also roots of P , the Claim follows. □

If P has a root other than 1, then applying the above Claim to a root of P having the largest absolute value yields a contradiction. Hence, P is a constant polynomial, or the only root of P is equal to 1.

If P is a constant, then P is equal to 0 or 1. If P is a nonconstant polynomial, then P is equal to $c(x-1)^n$ for some $c \in \mathbb{C} \setminus \{0\}$, and an integer $n \geq 1$. Using the hypothesis, it follows that $c = 1$.

Observing that the polynomials $0, 1, (x-1)^n$ satisfy the given condition, we conclude that there are all the required polynomials. ■

Exercise 8.4 (Problem 4.12, Putnam training problems by Miguel A. Lerma).

Does there exist a polynomial $f(x)$ satisfying

$$xf(x-1) = (x+1)f(x)?$$

Walkthrough —

(a)

Solution 47. Note that the zero polynomial is the only constant polynomial which satisfies the given condition.

Suppose there exists a nonconstant polynomial $f(x)$ satisfying the given condition.

Claim — Any root of $f(x)$ is an integer.

Proof of the Claim. On the contrary, let us assume that α is root of $f(x)$, and that α is not an integer. Note that for any root β of $f(x)$ in $\mathbb{C} \setminus \mathbb{Z}$, the element $\beta - 1$ is also a root of $f(x)$ lying in $\mathbb{C} \setminus \mathbb{Z}$. It follows that for any integer $k \geq 1$, the element $\alpha - k$ is also a root of $f(x)$. Taking k to be an integer larger than $|\alpha| + \sum_{\gamma \in \mathbb{C}, f(\gamma)=0} |\gamma|$, we obtain

$$|\alpha - k| \geq k - |\alpha| > |\gamma|,$$

and hence $f(x)$ has a root having absolute value larger than that of any of its roots, which is impossible. $\square$

A similar argument can be used to prove that $f(x)$ does not vanish at any negative integer. Moreover, noting that if $f(x)$ vanishes at an integer $n \geq -1$, then $f(x)$ also vanishes at $n+1 \geq -1$, it can be proved using a similar argument that the roots of $f(x)$ are ≤ -2 . This contradicts the assumption that there exists a nonconstant polynomial satisfying the given condition. Consequently, the zero polynomial is the only polynomial that satisfies the given condition. $\blacksquare$

Example 8.5 (Problem 2 of the Problem session for October 28, Fall 2020, Putnam Club). Find all polynomials $P(x)$ satisfying

$$xP(x-1) = (x-20)P(x).$$

Solution 48. Note that the zero polynomial is the only constant polynomial which satisfies the given condition.

Suppose $P(x)$ is a nonconstant polynomial satisfying the given condition.

Claim — Any root of $P(x)$ is an integer.

Proof of the Claim. On the contrary, let us assume that α is root of $P(x)$, and that α is not an integer. Note that for any root β of $P(x)$ in $\mathbb{C} \setminus \mathbb{Z}$, the element $\beta - 1$ is also a root of $P(x)$ lying in $\mathbb{C} \setminus \mathbb{Z}$. It follows that for any integer $k \geq 1$, the element $\alpha - k$ is also a root of $P(x)$. Taking k to be an integer larger than $|\alpha| + \sum_{\gamma \in \mathbb{C}, P(\gamma)=0} |\gamma|$, we obtain

$$|\alpha - k| \geq k - |\alpha| > |\gamma|,$$

and hence $P(x)$ has a root having absolute value larger than that of any of its roots, which is impossible. $\square$

A similar argument can be used to prove that no root of $P(x)$ is ≤ -1 or ≥ 20 . In other words, the roots of $P(x)$ are among $0, 1, 2, \dots, 19$. The given condition implies that if $0 \leq i \leq 19$ is a root of $P(x)$, then all of the integers $0, 1, 2, \dots, 19$ are roots of $P(x)$. Hence, $P(x)$ is equal to $cx^{n_0}(x-1)^{n_1} \dots (x-19)^{n_{19}}$ for some $c \in \mathbb{C}$, and positive integers $n_0, n_1, \dots, n_{19}$. Using the condition $xP(x-1) = (x-20)P(x)$, it follows that the integers $n_0, \dots, n_{19}$ are equal to 1. This shows that $P(x)$ is equal to $cx(x-1)(x-2) \dots (x-20)$. Moreover, any polynomial of this form satisfies the given condition. Hence, the polynomials satisfying the given condition are precisely of the above form. $\blacksquare$

Example 8.6 (India INMO 2018 P4). Find all polynomials $P(x)$ with real coefficients such that $P(x^2 + x + 1)$ divides $P(x^3 - 1)$.

Walkthrough —

- (a) Show that if α is a root of $P(x)$, then $P(x)$ vanishes at $(\beta_1 - 1)\alpha$ and $(\beta_2 - 1)\alpha$, where β_1, β_2 are the roots of $x^2 + x + 1 = \alpha$.
- (b) If α is nonzero, then show that one of $(\beta_1 - 1)\alpha$ and $(\beta_2 - 1)\alpha$ is larger than α in absolute value.

Solution 49. Let us establish the following claim.

Claim — Let α denote a nonzero root of $P(x)$ in $\mathbb{C}$. Then for some $\beta \in \mathbb{C}$ satisfying $\beta^2 + \beta + 1 = \alpha$, the element $(\beta - 1)\alpha$ is a root of $P(x)$ and is larger than α in absolute value.

Proof of the Claim. Let β_1, β_2 denote the roots of $x^2 + x + 1 = \alpha$ in $\mathbb{C}$. Since $P(x^2 + x + 1)$ divides the polynomial $P(x^3 - 1)$, it follows that for any $1 \leq i \leq 2$,

$$\beta_i^3 - 1 = (\beta_i - 1)(\beta_i^2 + \beta_i + 1) = (\beta_i - 1)\alpha$$

is a root of $P(x)$. Noting that

$$\begin{aligned} |\beta_1 - 1| + |\beta_2 - 1| &\geq |\beta_1 + \beta_2 - 2| \\ &= |-1 - 2| \\ &> 2, \end{aligned}$$

we obtain that at least one of $\beta_1 - 1, \beta_2 - 1$ has absolute value larger than 1. Consequently, $|(\beta_i - 1)\alpha| > |\alpha|$ holds for some $1 \leq i \leq 2$. This proves the Claim. $\square$

If $P(x)$ has a nonzero root in $\mathbb{C}$, then applying the above Claim to a root of $P(x)$ of largest absolute value, we would obtain a contradiction. Hence, $P(x)$ is equal to cx^n for some $c \in \mathbb{R}$ and an integer $n \geq 0$. Moreover, any polynomial of this form satisfies the given condition. Consequently, these are all the polynomials satisfying the given condition. $\blacksquare$

Remark. Note that proving $|(\beta_1 - 1)(\beta_2 - 1)| > 1$, in order to conclude that at least one of $\beta_1 - 1, \beta_2 - 1$ has absolute value larger than 1, does not seem to work. Moreover, $|(\beta_1 - 1)(\beta_2 - 1)|$ is smaller than $\frac{1}{2}(|\beta_1 - 1| + |\beta_2 - 1|)$. **Unsurprisingly**, a lower bound for the bigger quantity can easily be obtained.

Exercise 8.7 (IMOSL 1979, AoPS, Bulgaria). Find all polynomials $f(x)$ with real coefficients satisfying

$$f(x)f(2x^2) = f(2x^3 + x).$$

Walkthrough —

(a)

Solution 50. Note that $0, 1$ are the only constant polynomials satisfying the given condition.

Let $f(x)$ be a nonconstant polynomial with real coefficients satisfying the given condition.

Claim — The roots of $f(x)$ in $\mathbb{C}$ are of absolute value at most 1.

Proof of the Claim. If α is a root of $f(x)$ in $\mathbb{C}$, then $2\alpha^3 + \alpha$ is a root of $f(x)$. If $|\alpha| > 1$, then

$$|2\alpha^3 + \alpha| \geq 2|\alpha|^3 - |\alpha| > |\alpha|$$

holds. If some root of $f(x)$ has absolute value larger than 1, then taking α to be a root of $f(x)$ with largest absolute value, we would obtain a contradiction. This proves the Claim. $\square$

Claim — The equality $f(0) = 1$ holds.

Proof of the Claim. Substituting $x = 0$, it follows that $f(0)^2 = f(0)$, and hence $f(0) = 0$ or $f(0) = 1$.

On the contrary, suppose $f(0) = 0$ holds. Write $f(x) = x^k g(x)$ where k is a positive integer, and $g(x)$ is a polynomial with real coefficients with $g(0) \neq 0$. The given condition translates to

$$x^k (2x^2)^k g(x) g(2x^2) = (2x^3 + x)^k g(2x^3 + x),$$

which yields

$$(2x^2)^k g(x) g(2x^2) = (2x^2 + 1)^k g(2x^3 + x).$$

Substituting $x = 0$, we obtain $g(0) = 0$, which is impossible. This proves the Claim. $\square$

By the above Claim, the product of the absolute values of the roots of $f(x)$ is equal to 1. By the first Claim, these absolute values are at most 1. It follows that the roots of $f(x)$ are of absolute value 1.

Let α be a root of $f(x)$. Note that $2\alpha^3 + \alpha$ is also a root of $f(x)$, and we have that

$$|\alpha| = |2\alpha^3 + \alpha| = 1,$$

which yields $|2\alpha^2 + 1| = |\alpha| = 1$. This gives

$$(2\alpha^2 + 1)(2\bar{\alpha}^2 + 1) = 1.$$

Combining it with $|\alpha| = 1$, we obtain

$$\alpha^2 + \bar{\alpha}^2 = -2.$$

Since $|\alpha| = 1$, it follows that $\alpha^2 = -1$, and hence, $\alpha = i$ or $\alpha = -i$. So, the polynomial $f(x)$ is equal to $c(x+i)^a(x-i)^b$ for some $c \in \mathbb{C} \setminus \{0\}$, and some nonnegative integers a, b . Since $f(x)$ has real coefficients, it follows that $a = b$, and c lies in $\mathbb{R}$. This gives that $f(x) = c(x^2 + 1)^a$. Using $f(0) = 1$, we obtain $c = 1$, and hence $f(x) = (x^2 + 1)^a$.

Note that if $g(x)$ denotes the polynomial $(x^2 + 1)^k$, where k is a positive integer, then

$$\begin{aligned} g(x)g(2x^2) &= ((x^2 + 1)((2x^2)^2 + 1))^k \\ &= ((4x^6 + 4x^4 + x^2 + 1))^k \\ &= (((2x^3 + x)^2 + 1))^k \\ &= g(2x^3 + x). \end{aligned}$$

We conclude that the polynomials satisfying the given condition are precisely the constant polynomial 0, 1, and the polynomials of the form $(x^2 + 1)^k$ for some positive integer k . $\blacksquare$

Theorem 5 (Cauchy)

Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial with complex coefficients such that $a_n \neq 0$ and $n \geq 1$. Then any root $\alpha \in \mathbb{C}$ of $P(x)$ satisfies

$$|\alpha| < 1 + \frac{\max\{|a_{n-1}|, |a_{n-2}|, \dots, |a_1|, |a_0|\}}{|a_n|}.$$

Moreover, if $k \geq 0$ is an integer such that

$$a_{n-1} = a_{n-2} = \cdots = a_{n-k} = 0,$$

then any root $\alpha \in \mathbb{C}$ of $P(x)$ satisfies

$$|\alpha| \leq 1 + \left(\frac{\max\{|a_{n-k-1}|, |a_{n-k-2}|, \dots, |a_1|, |a_0|\}}{|a_n|} \right)^{\frac{1}{k+1}}.$$

Proof. Let α be a root of P in $\mathbb{C}$. Write

$$M = \max\{|a_{n-1}|, |a_{n-2}|, \dots, |a_1|, |a_0|\}.$$

Note that

$$a_n \alpha^n = -(a_{n-1} \alpha^{n-1} + a_{n-2} \alpha^{n-2} + \cdots + a_1 \alpha + a_0)$$

holds, which implies

$$|a_n| |\alpha|^n \leq M(|\alpha|^{n-1} + |\alpha|^{n-2} + \cdots + |\alpha| + 1).$$

It suffices to consider the case $|\alpha| > 1$, since otherwise the desired inequality follows. Using

$$|\alpha|^n > (|\alpha| - 1) (|\alpha|^{n-1} + |\alpha|^{n-2} + \cdots + |\alpha| + 1),$$

we obtain

$$(|\alpha| - 1) |a_n| < M,$$

which gives the first desired inequality.

Moreover, if $k \geq 0$ is an integer such that

$$a_{n-1} = a_{n-2} = \cdots = a_{n-k} = 0,$$

then we have

$$|a_n| |\alpha|^n \leq M(|\alpha|^{n-k-1} + |\alpha|^{n-k-2} + \cdots + |\alpha| + 1).$$

Again, it suffice to consider the case $|\alpha| > 1$. Note that for any real number $r > 1$, writing $r = 1 + s$, we obtain

$$\begin{aligned}
 r^n &= (1 + s)^n \\
 &= \sum_{i=0}^n \binom{n}{i} s^i \\
 &> \sum_{i=k+1}^n \binom{n}{i} s^i \\
 &\geq \sum_{i=k+1}^n \binom{n-k}{i-k} s^i \\
 &= s^k \sum_{j=1}^{n-k} \binom{n-k}{j} s^j \\
 &= s^k ((1 + s)^{n-k} - 1) \\
 &= (r - 1)^k (r^{n-k} - 1) \\
 &= (r - 1)^{k+1} (r^{n-k-1} + r^{n-k-2} + \dots + r + 1).
 \end{aligned}$$

The above implies that

$$|a_n|(|\alpha| - 1)^{k+1} < M,$$

which gives the second desired inequality. $\square$

Exercise 8.8 (Balkan Math Olympiad 1989 P2, AoPS, Yugoslavia). Let $\overline{a_n a_{n-1} \dots a_1 a_0}$ be the decimal representation of a prime number such that $n > 1$ and $a_n > 1$. Prove that the polynomial

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

cannot be expressed as a product of two non-constant polynomials with integer coefficients.

Solution 51. On the contrary, let us assume that

$$P(x) = Q(x)R(x)$$

for some non-constant polynomials $Q(x), R(x)$ with integer coefficients. Since $P(10)$ is a prime, by reordering Q and R if necessary, we may assume without loss of generality that $|Q(10)| = 1$. Using Cauchy's theorem on the size of roots of polynomials (see Theorem 5) and that $a_n \geq 2$, it follows that the roots of $P(x)$ have absolute value less than

$$1 + \frac{9}{2}.$$

Let $r_1, \dots, r_k$ denote the roots of $Q(x)$ in $\mathbb{C}$, counted with multiplicity. Then we have

$$|Q(10)| = |a| \prod_{i=1}^k |10 - r_i|,$$

where a is the leading coefficient of $Q(x)$. Using the triangle inequality, we obtain

$$|Q(10)| \geq |a| \prod_{i=1}^k (10 - |r_i|) > |a| \left(10 - \frac{11}{2}\right)^k > 1,$$

which is a contradiction. Hence the polynomial $P(x)$ cannot be expressed as a product of two non-constant polynomials with integer coefficients. ■

§9 Roots of unity

Some of the problems have been borrowed from Yufei Zhao's handout on polynomials.

Example 9.1 (India Pre-RMO 2012 P17). Let x_1, x_2, x_3 be the roots of the equation $x^3 + 3x + 5 = 0$. What is the value of the expression

$$\left(x_1 + \frac{1}{x_1}\right) \left(x_2 + \frac{1}{x_2}\right) \left(x_3 + \frac{1}{x_3}\right)?$$

See also ??, USAMO 2014 P1.

Solution 52. Let $P(x)$ denote the polynomial $x^3 + 3x + 5$. Note that

$$\begin{aligned} & \left(x_1 + \frac{1}{x_1}\right) \left(x_2 + \frac{1}{x_2}\right) \left(x_3 + \frac{1}{x_3}\right) \\ &= \frac{1}{x_1 x_2 x_3} (x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1) \\ &= \frac{1}{x_1 x_2 x_3} (x_1 + i)(x_2 + i)(x_3 + i)(x_1 - i)(x_2 - i)(x_3 - i) \\ &= \frac{1}{x_1 x_2 x_3} P(-i)P(i) \\ &= \frac{1}{-5} |P(i)|^2 \\ &= \frac{1}{-5} |5 - 2i|^2 \\ &= -\frac{29}{5}. \end{aligned}$$

■

Remark. The above argument is elegant and quite useful. One could have also argued that

$$\begin{aligned}
 & \left(x_1 + \frac{1}{x_1}\right) \left(x_2 + \frac{1}{x_2}\right) \left(x_3 + \frac{1}{x_3}\right) \\
 &= \left(x_1x_2 + \frac{x_1}{x_2} + \frac{x_2}{x_1} + \frac{1}{x_1x_2}\right) \left(x_3 + \frac{1}{x_3}\right) \\
 &= x_1x_2x_3 + \frac{x_1x_3}{x_2} + \frac{x_2x_3}{x_1} + \frac{x_3}{x_1x_2} + \frac{x_1x_2}{x_3} + \frac{x_1}{x_2x_3} + \frac{x_2}{x_3x_1} + \frac{1}{x_1x_2x_3} \\
 &= x_1x_2x_3 + \frac{1}{x_1x_2x_3} + \frac{x_1x_2}{x_3} + \frac{x_2x_3}{x_1} + \frac{x_3x_1}{x_2} + \frac{x_1}{x_2x_3} + \frac{x_2}{x_3x_1} + \frac{x_3}{x_1x_2} \\
 &= -5 - \frac{1}{5} + \frac{1}{x_1x_2x_3} (x_1^2x_2^2 + x_2^2x_3^2 + x_3^2x_1^2) + \frac{1}{x_1x_2x_3} (x_1^2 + x_2^2 + x_3^2) \\
 &= -5 - \frac{1}{5} - \frac{1}{5} ((x_1x_2 + x_2x_3 + x_3x_1)^2 - 2x_1x_2x_3(x_1 + x_2 + x_3)) \\
 &\quad - \frac{1}{5} ((x_1 + x_2 + x_3)^2 - 2(x_1x_2 + x_2x_3 + x_3x_1)) \\
 &= -5 - \frac{1}{5} - \frac{1}{5} (3^2) - \frac{1}{5} (-2 \cdot 3) \\
 &= -5 - \frac{1}{5} - \frac{9}{5} + \frac{6}{5} \\
 &= -5 - \frac{4}{5} \\
 &= -\frac{29}{5}.
 \end{aligned}$$

Note that this way of arguing would get complicated if we had a higher degree polynomial to start with.

Exercise 9.2 (USAMO 2014 P1, AoPS). Let a, b, c, d be real numbers such that $b - d \geq 5$ and all zeros x_1, x_2, x_3, x_4 of the polynomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ are real. Find the smallest value the product

$$(x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1)$$

can take.

See also Example 9.1, [India Pre-RMO 2012 P17](#).

Walkthrough —

(a)

Solution 53. Note that

$$\begin{aligned}
 (x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1) &= P(i)P(-i) \\
 &= (1 - b + d)^2 + (a - c)^2
 \end{aligned}$$

$$\begin{aligned}
&= (b - d - 1)^2 + (a - c)^2 \\
&\geq 16.
\end{aligned}$$

Taking $a = c$ and $b = 5, d = 0$, we obtain

$$(x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1) = 16.$$

Hence, the smallest value the product $(x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1)$ can take is equal to 16. ■

Exercise 9.3 (USAMO 1976 P5, AoPS). If $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ are all polynomials such that

$$P(x^5) + xQ(x^5) + x^2R(x^5) = (x^4 + x^3 + x^2 + x + 1)S(x),$$

prove that $x - 1$ is a factor of $P(x)$.

Summary — The primitive 5-th roots of unity can be used to show that $P(1) = 0$.

Walkthrough —

- (a) Substituting primitive 5-th roots of unity (that is, the 5-th roots of unity other than 1) for x , yields several linear equations in $P(1), Q(1), R(1)$.
- (b) Can $Q(1)$ and $R(1)$ be eliminated to obtain that $P(1) = 0$?

Solution 54. Denote the 5-th root of unity $\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$ by ζ . Substituting $x = \zeta, \zeta^2, \zeta^3$, we obtain

$$\begin{aligned}
P(1) + \zeta Q(1) + \zeta^2 R(1) &= 0, \\
P(1) + \zeta^2 Q(1) + \zeta^4 R(1) &= 0, \\
P(1) + \zeta^3 Q(1) + \zeta^6 R(1) &= 0.
\end{aligned}$$

Eliminating $R(1)$ from the first two equations yields

$$(1 - \zeta^2)P(1) + \zeta^2(1 - \zeta)Q(1) = 0,$$

and eliminating $R(1)$ from the last two equations yields

$$(1 - \zeta^2)P(1) + \zeta^3(1 - \zeta)Q(1) = 0.$$

Eliminating $Q(1)$ from the above two equations, we obtain $(1 - \zeta)P(1) = 0$, which gives $P(1) = 0$. This shows that $x - 1$ is a factor of $P(x)$. ■

Exercise 9.4 (Leningrad Math Olympiad 1991). A finite sequence $a_1, a_2, \dots, a_n$ is called p -balanced if any sum of the form

$$a_k + a_{k+p} + a_{k+2p} + \dots$$

is the same for any $k = 1, 2, 3, \dots, p$. For instance the sequence

$$a_1 = 1, a_2 = 2, a_3 = 3, a_4 = 4, a_5 = 3, a_6 = 2$$

is a 3-balanced. Prove that if a sequence with 50 members is p -balanced for $p = 3, 5, 7, 11, 13, 17$, then all its members are equal zero.

Summary — Consider the polynomial $\sum_{i=1}^{50} a_i x^i$.

Walkthrough —

- (a) Show that the polynomial vanishes at any p -th root of unity, other than 1, for $p \in \{3, 5, 7, 11, 13, 17\}$.
- (b) How many such roots of unity are there in total?

Solution 55. Let $a_1, a_2, \dots, a_{50}$ be a sequence of complex numbers. Assume that it is p -balanced for $p \in \{3, 5, 7, 11, 13, 17\}$. For an integer $n \geq 1$, denote the root of unity $\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$ by ζ_n . Let $P(x)$ denote the polynomial $\sum_{i=1}^n a_i x^i$.

Let $3 \leq p \leq 17$ be a prime. Since $a_1, a_2, \dots, a_{50}$ is p -balanced, for any $1 \leq \ell < p$, we obtain

$$\begin{aligned} P(\zeta_p^\ell) &= \sum_{k=1}^p (a_k + a_{k+p} + \dots) \zeta_p^{k\ell} \\ &= (a_1 + a_{1+p} + \dots) \sum_{k=1}^p \zeta_p^{k\ell} \\ &= 0, \end{aligned}$$

where the final equality follows since $\zeta_p^\ell \neq 1$. This shows that the polynomial $P(x)$ vanishes at the elements of the set

$$\cup_{p \in \{3, 5, 7, 11, 13, 17\}} \{\zeta_p^\ell \mid 1 \leq \ell < p\},$$

which contains

$$\sum_{p \in \{3, 5, 7, 11, 13, 17\}} (p-1) = 2 + 4 + 6 + 10 + 12 + 16 = 50$$

elements. Moreover, $P(x)$ also vanishes at 0. Note that $P(x)$ is a polynomial of degree 50, and it has at least 51 zeroes. This gives that $P(x) = 0$, and hence, the terms of the sequence $a_1, a_2, \dots, a_{50}$ are all equal to zero. ■

Exercise 9.5 (USAMO 1977 P1, AoPS). Determine all pairs of positive integers (m, n) such that $1 + x^n + x^{2n} + \cdots + x^{mn}$ is divisible by $1 + x + x^2 + \cdots + x^m$.

Solution 56. Let μ_{m+1} denote the set of all $(m+1)$ -th roots of unity in $\mathbb{C}$. Note that

$$1 + x + x^2 + \cdots + x^m = \frac{x^{m+1} - 1}{x - 1} = \prod_{\zeta \in \mu_{m+1} \setminus \{1\}} (x - \zeta).$$

Moreover, we have

$$1 + x^n + x^{2n} + \cdots + x^{mn} = \prod_{\xi \in \mu_{m+1} \setminus \{1\}} (x^n - \xi).$$

If $m+1, n$ are relatively prime, then the map $x \mapsto x^n$ is a bijection on μ_{m+1} , and it fixes the point 1, and hence,

$$1 + x^n + x^{2n} + \cdots + x^{mn} = \prod_{\zeta \in \mu_{m+1} \setminus \{1\}} (x^n - \zeta^n),$$

which shows that $1 + x + x^2 + \cdots + x^m$ divides $1 + x^n + x^{2n} + \cdots + x^{mn}$.

Let d denote the greatest common divisor of $m+1$ and n . Assume that $d > 1$. Then, there exists a $(m+1)$ -th root of unity ζ such that $\zeta^n = 1$ and $\zeta \neq 1$. Indeed, let ω be a primitive $(m+1)$ -th root of unity. Then, $\omega^{(m+1)/d}$ is such a root of unity. Note that

$$1 + x^n + x^{2n} + \cdots + x^{mn}$$

does not vanish at $x = \zeta$, and hence $1 + x + x^2 + \cdots + x^m$ does not divide $1 + x^n + x^{2n} + \cdots + x^{mn}$. This shows that $1 + x + x^2 + \cdots + x^m$ divides $1 + x^n + x^{2n} + \cdots + x^{mn}$ if and only if $m+1, n$ are relatively prime. ■

Example 9.6. Show that any function $f : \mathbb{R} \rightarrow \mathbb{R}$ can be written as a sum of an even function and an odd function.

Solution 57. Consider the functions $f_1, f_{-1} : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f_\varepsilon(x) = \frac{1}{2}(f(x) + \varepsilon f(\varepsilon x)), \quad \text{for all } x \in \mathbb{R},$$

where $\varepsilon \in \{1, -1\}$. Note that f_1 is an even function, f_{-1} is an odd function, and that $f = f_1 + f_{-1}$ holds. ■

Remark. It seems from the above solution that we “knew” the functions f_1, f_{-1} beforehand! Here is another solution to the above problem.

Let us assume that f can be expressed as the sum of an even function g and an odd function h . Note that

$$f(x) = g(x) + h(x),$$

$$\begin{aligned} f(-x) &= g(-x) + h(-x) \\ &= g(x) - h(x) \end{aligned}$$

hold for all $x \in \mathbb{R}$. Combining the above, we obtain

$$\begin{aligned} g(x) &= \frac{1}{2}(f(x) + f(-x)), \\ h(x) &= \frac{1}{2}(f(x) - f(-x)) \end{aligned}$$

for all $x \in \mathbb{R}$.

Note that this is **NOT a proof**. We have observed that

$$\begin{aligned} g(x) &= \frac{1}{2}(f(x) + f(-x)), \\ h(x) &= \frac{1}{2}(f(x) - f(-x)) \end{aligned}$$

holds **relying on the assumption that** g is an even function, and h is an odd function satisfying $f = g + h$. This observation is **made under the hypothesis** that $f = g + h$ where g (resp. h) is even (resp. odd), and **cannot be a proof of the hypothesis!** However, this observation can be used to arrive at the above proof, by defining the functions f_1, f_{-1} as above, and showing that they have the required properties.

Example 9.7. Let ω denote the root of unity $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$. Show that any function $f : \mathbb{C} \rightarrow \mathbb{C}$ can be written as a sum of three functions $f_0, f_1, f_2 : \mathbb{C} \rightarrow \mathbb{C}$ satisfying

$$f_i(\omega z) = \bar{\omega}^i z \quad \text{for all } z \in \mathbb{C}.$$

Walkthrough — Assume that the function f can be expressed as stated, and then determine some of the properties of the functions f_0, f_1, f_2 , as done in the above remark. What remains left to do?

Example 9.8. Let $P(x)$ be a monic polynomial with integer coefficients such that all its zeroes lie on the unit circle. Show that all the zeroes of $P(x)$ are roots of unity, that is, $P(x)$ divides $(x^n - 1)^k$ for some positive integers n, k .

The author has learnt the following argument from **Mainak Ghosh**, during the **INMO Training Camp 2025**, held at IISER Bhopal.

Walkthrough —

- (a) Use the fundamental theorem of symmetric polynomials, to prove the following claim.

Claim — Let $f(x)$ be a monic polynomial of degree n with integer coefficients. Let $\alpha_1, \dots, \alpha_n$ denote its roots, counting multiplicities. Then for any integer $k \geq 1$, there is a monic polynomial of degree n with integer coefficients, having $\alpha_1^k, \alpha_2^k, \dots, \alpha_n^k$ as its roots.

- (b) Applying the Claim, for each integer $k \geq 1$, obtain a monic polynomial $P_k(x)$ with integer coefficients, having degree same as that of $P(x)$, whose roots, counted with multiplicities, are the k -th powers of the roots of $P(x)$.
- (c) Note that the polynomials $P_1(x), P_2(x), \dots$ are of the same degree, and the absolute values of the coefficients of any of them are bounded from the above by suitable binomial coefficients, which is smaller than 2^n , where n denotes the degree of $P(x)$. Since these polynomials have integer coefficients, by the pigeonhole principle, it follows that there is a positive integer k_0 such that $P_k(x) = P_{k_0}(x)$ holds for infinitely many positive integers k .
- (d) Enumerate the roots of $P(x)$, and matching the roots of $P_k(x)$ with those of $P_{k_0}(x)$ (for suitable k 's), we obtain a permutation of n letters. By the pigeonhole principle, infinitely many k 's yield the same permutation, which implies that there are positive integers $k \neq \ell$ such that for any root of $P(x)$, its k -th and the ℓ -th powers are equal.

Solution 58. ■

§10 Complex numbers

Exercise 10.1 (Moldova National Olympiad 2001 Grade 11 Day 2 P6, AoPS).

For a positive integer n , denote $A_n = \{(x, y) \in \mathbb{Z}^2 \mid x^2 + xy + y^2 = n\}$.

- (a) Prove that the set A_n is finite.
- (b) Prove that the number of elements of A_n is divisible by 6 for all $n \geq 1$.
- (c) For which n is the number of elements of A_n divisible by 12?

Remark. The following walkthrough may be skipped if one does not have an exposure to the notion of **groups**.

Walkthrough —

- (a) Note that $x^2 + xy + y^2 \geq (x + \frac{y}{2})^2 + \frac{3}{4}y^2$ holds.
- (b) For part (b), consider the transformations

$$(x, y) \mapsto (y, -x - y), \quad (x, y) \mapsto (y, x).$$

Show that these transformations^a map A_n to itself and generate a group

of order 6 acting on A_n without fixed points.

(c) For part (c), consider the transformations

$$(x, y) \mapsto (x, y), \quad (x, y) \mapsto (-x, -y).$$

Show that these transformations map A_n to itself, and provides an action of the group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ on A_n . Determine the fixed points of this action.

^aNote that

$$(x - y\omega)\omega = x\omega - y\omega^2 = y - (x + y)\omega$$

holds.

Solution 59. Note that any $(x, y) \in A_n$ satisfies

$$n = x^2 + xy + y^2 = \frac{1}{2} \left((x + y)^2 + x^2 + y^2 \right),$$

which implies that $|x|, |y| \leq \sqrt{2n}$. Thus, there are only finitely many pairs (x, y) of integers satisfying $x^2 + xy + y^2 = n$. This proves part (a).

To establish part (b), assume that n is a positive integer. For any $(x, y) \in A_n$, consider the pairs

$$(x, y), (-x - y, x), (y, -x - y),$$

and note that they all belong to A_n . Indeed,

$$\begin{aligned} (-x - y)^2 + (-x - y)x + x^2 &= x^2 + xy + y^2, \\ y^2 + y(-x - y) + (-x - y)^2 &= x^2 + xy + y^2. \end{aligned}$$

Further, if any two of these three pairs were equal, then we would have $x = y = 0$, which is impossible since $n \geq 1$. Thus, the three pairs above are distinct. Hence, the elements of A_n can be grouped into sets of three, proving that $|A_n|$ is divisible by 3. Similarly, for any $(x, y) \in A_n$, the pair (y, x) also belongs to A_n , and is distinct from (x, y) unless $n/3$ is a perfect square. Moreover, for any $(x, y) \in A_n$, the pair $(-y, -x)$ also belongs to A_n , and is distinct from (x, y) unless n is a perfect square. Thus, the elements of A_n can be grouped into sets of two, proving that $|A_n|$ is even. Combining these results, we conclude that $|A_n|$ is divisible by 6.

Finally, to determine when $|A_n|$ is divisible by 12, it suffices to determine when $|A_n|$ is divisible by 4. For any $(x, y) \in A_n$, consider the pairs

$$(x, y), (y, x), (-x, -y), (-y, -x),$$

and note that they all belong to A_n . Moreover, if any two of these four pairs were equal, then we would have either $x = y$ with $n = 3x^2$, or $x = -y$ with $n = x^2$. Therefore, if $n/3$ is not a perfect square and n is not a perfect square, then the four pairs above are distinct, and consequently $|A_n|$ is divisible by 4. Moreover, if $n = 3k^2$ holds for some integer k , then the pairs $(k, k), (-k, -k)$

belongs to A_n , and the remaining elements of A_n can be grouped into sets of four, showing that $|A_n| \equiv 2 \pmod{4}$. Similarly, if $n = k^2$ holds for some integer k , then the pairs $(k, -k), (-k, k)$ belongs to A_n , and the remaining elements of A_n can be grouped into sets of four, showing that $|A_n| \equiv 2 \pmod{4}$. Thus, $|A_n|$ is divisible by 4 if and only if $n/3$ is not a perfect square and n is not a perfect square. ■

Exercise 10.2 (Putnam 2018 B2, AoPS). Let n be a positive integer, and let $f_n(z) = n + (n-1)z + (n-2)z^2 + \cdots + 2z^{n-2} + z^{n-1}$. Show that all the roots of $f_n(z)$ lie outside the closed unit disk $\{z \in \mathbb{C} : |z| \leq 1\}$.

Walkthrough —

(a)

Solution 60. Let us assume that there exists a root z of f_n such that $|z| \leq 1$. Note that

$$(1-z)f_n(z) = n - z - z^2 - \cdots - z^{n-1} - z^n.$$

Using the triangle inequality, we have

$$\begin{aligned} 0 &= |(1-z)f_n(z)| &&= |n - z - z^2 - \cdots - z^{n-1} - z^n| \\ &\geq n - |z + z^2 + \cdots + z^{n-1} + z^n| \\ &\geq n - |z| - |z^2| - \cdots - |z^{n-1}| - |z^n| \\ &> n - 1 - 1 - \cdots - 1 - 1 \\ &= 0. \end{aligned}$$

This shows that $z, z^2, \dots, z^n$ have absolute value 1, and have the same argument, and hence $z = 1$. However, $f_n(1) = n + (n-1) + (n-2) + \cdots + 2 + 1$, which is nonzero. This is a contradiction. Hence, all the roots of f_n lie outside the closed unit disk. ■

Solution 61. Let $g_n(z)$ denote the polynomial $z^n + z^{n-1} + \cdots + z + 1$. Note that

$$g'_n(z) = nz^{n-1} + (n-1)z^{n-2} + \cdots + 2z + 1 = z^{n-1}f_n\left(\frac{1}{z}\right).$$

Observe that the roots of the polynomials g'_n, f_n are nonzero, and a nonzero complex number z is a root of f_n if and only if $\frac{1}{z}$ is a root of g'_n . By the Gauss–Lucas theorem (Theorem 3), all the roots of g'_n lie in the convex hull of the set of roots of g_n . Note that the roots of g_n are precisely the n -th roots of unity other than 1. Since g_n has no multiple roots, it follows that g'_n does not vanish at any of the roots of g_n . Thus, all the roots of g'_n lie in the interior of the convex hull of the set of roots of g_n , and consequently, the roots of g'_n lie in the open unit disk. Therefore, all the roots of f_n lie outside the closed unit disk. ■

§11 Crossing the x -axis

Here are a few problems from [this notes](#), and [this one](#).

Example 11.1. Suppose $P(x)$ is a polynomial with real coefficients such that $P(x) = x$ has no real solution. Show that $P(P(x)) = x$ has no real solutions.

Walkthrough —

(a)

Solution 62. Since $x \mapsto P(x)$ defines a continuous map from $\mathbb{R} \rightarrow \mathbb{R}$, by the intermediate value theorem, it follows that $P(x) > x$ holds for all $x \in \mathbb{R}$ or $P(x) < x$ holds for all $x \in \mathbb{R}$. If $P(x) > x$ holds for all $x \in \mathbb{R}$, then $P(P(x)) > P(x) > x$ holds for all $x \in \mathbb{R}$, and hence $P(P(x)) = x$ has no real solutions. Similarly, if $P(x) < x$ holds for all $x \in \mathbb{R}$, then $P(P(x)) = x$ has no real solutions. ■

Exercise 11.2 (All-Russian Mathematical Olympiad 2013 Grade 9 Day 1 P1, AoPS, by I. Bogdanov). Given three distinct real numbers a , b , and c , show that at least two of the three following equations

$$(x - a)(x - b) = x - c,$$

$$(x - b)(x - c) = x - a,$$

$$(x - c)(x - a) = x - b$$

have real solutions.

Walkthrough —

(a)

Solution 63. Without loss of generality, let us consider the case that the equations

$$(x - a)(x - b) = x - c, (x - b)(x - c) = x - a$$

do not any real root. It follows that their discriminants are negative, that is,

$$(a + b + 1)^2 - 4(ab + c), (b + c + 1)^2 - 4(bc + a)$$

are negative. Note that

$$(a + b + 1)^2 - 4(ab + c) = (a - b)^2 + 1 + 2(a + b - 2c)$$

$$= (b - a + 1)^2 + 4(a - c),$$

$$(b + c + 1)^2 - 4(bc + a) = (b - c)^2 + 1 + 2(b + c - 2a)$$

$$= (b - c + 1)^2 + 4(c - a).$$

Since a and c are distinct real numbers, one of $a - c$ and $c - a$ is positive. Thus, the discriminant of one of the equations

$$(x - a)(x - b) = x - c, (x - b)(x - c) = x - a$$

is non-negative, which is a contradiction. Therefore, at least two of the three equations have real solutions. ■

Solution 64. Let $f(x)$, $g(x)$, and $h(x)$ denote the polynomials defined by

$$f(x) = (x - a)(x - b) - (x - c),$$

$$g(x) = (x - b)(x - c) - (x - a),$$

$$h(x) = (x - c)(x - a) - (x - b).$$

Note that

$$\begin{aligned} f(x) + g(x) &= (x - b)(2x - a - c) - (2x - a - c) \\ &= (2x - a - c)(x - b - 1). \end{aligned}$$

This implies that $f(x) + g(x)$ admits a real root. Hence, at least one of $f(x)$ and $g(x)$ is non-positive at some real number. If one of them admits a real root, then we are done. Otherwise, at least one of them is negative at some real number. Also note that if k is a real number satisfying $k \geq |a| + |b| + |c| + 2$, then

$$\begin{aligned} f(k + |a| + |b|) &= (k + |a| + |b| + a)(k + |a| + |b| + b) - (k + |a| + |b| + c) \\ &\geq k^2 - k - |a| - |b| - |c| \\ &\geq k - |a| - |b| - |c| \\ &> 0 \end{aligned}$$

holds, and similarly,

$$g(k + |a| + |b|) > 0$$

holds too. By the intermediate value theorem, at least one of $f(x)$ and $g(x)$ admits a real root. Interchanging the roles of f , g , and h , it follows that at least one of any two of the polynomials f , g , and h admits a real root. Therefore, at least two of the three equations have real solutions for x . ■

Exercise 11.3 (All-Russian Mathematical Olympiad 2007 Grade 8 Day 1 P1, AoPS, by O. Podlipsky). Given real numbers a , b , c , prove that at least one of three equations

$$x^2 + (a - b)x + (b - c) = 0,$$

$$x^2 + (b - c)x + (c - a) = 0,$$

$$x^2 + (c - a)x + (a - b) = 0$$

has a real root.

Walkthrough —**(a)**

Solution 65. Note that the sum of the discriminants of the quadratic polynomials

$$x^2 + (a - b)x + (b - c), x^2 + (b - c)x + (c - a), x^2 + (c - a)x + (a - b)$$

are equal to

$$\begin{aligned} & (a - b)^2 - 4(b - c) + (b - c)^2 - 4(c - a) + (c - a)^2 - 4(a - b) \\ &= (a - b)^2 + (b - c)^2 + (c - a)^2, \end{aligned}$$

which is non-negative, and hence, at least one of the discriminants is non-negative. This implies that at least one of the three equations has a real root. ■

Solution 66. Consider the three quadratic polynomials defined by

$$\begin{aligned} f(x) &= x^2 + (a - b)x + (b - c), \\ g(x) &= x^2 + (b - c)x + (c - a), \\ h(x) &= x^2 + (c - a)x + (a - b). \end{aligned}$$

Note that

$$f(0) + g(0) + h(0) = 0,$$

which implies that at least one of $f(0)$, $g(0)$, $h(0)$ is non-positive. Without loss of generality, assume that $f(0) \leq 0$. If $f(0)$ is zero, then $x = 0$ is a root of $f(x)$. If $f(0) < 0$, then noting that

$$\begin{aligned} & f(2|a - b| + 2|b - c| + 1) \\ &= \frac{1}{2}(2|a - b| + 2|b - c| + 1)^2 + (a - b)(2|a - b| + 2|b - c| + 1) \\ &\quad + \frac{1}{2}(2|a - b| + 2|b - c| + 1)^2 + (b - c) \\ &= \frac{1}{2}(2|a - b| + 2|b - c| + 1)(2|a - b| + 2|b - c| + 1 - 2(a - b)) \\ &\quad + \frac{1}{2}(2|a - b| + 2|b - c| + 1)^2 + (b - c) \\ &\geq 1 \end{aligned}$$

holds, it follows from the intermediate value theorem that $f(x)$ has a real root. ■

Exercise 11.4 (All-Russian Mathematical Olympiad 2007 Grade 9 Day 1 P1, AoPS, by S. Berlov). Let $f(x)$, $g(x)$ be monic quadratic polynomials with real coefficients such that the equations $f(g(x)) = 0$ and $g(f(x)) = 0$ have no real roots. Prove that at least one of the equations $f(f(x)) = 0$ and $g(g(x)) = 0$ has no real roots.

Walkthrough —

(a)

Solution 67. If one of the polynomials $f(x)$, $g(x)$ has no real roots, then the conclusion follows immediately. Thus, we may assume that both $f(x)$ and $g(x)$ have real roots. Denote the roots of $f(x)$ by α_1, α_2 , and the roots of $g(x)$ by β_1, β_2 . We need to show that none of α_1, α_2 lies in $f(\mathbb{R})$, or none of β_1, β_2 lies in $g(\mathbb{R})$.

Claim — If $\alpha_2 \leq \beta_2$, then none of α_1, α_2 lies in $f(\mathbb{R})$.

Proof of the Claim. Let us assume that $\alpha_1 \leq \beta_2$. If some of α_1, α_2 lies in $f(\mathbb{R})$, then noting that f is a monic quadratic polynomial, it follows from the intermediate value theorem that β_2 lies in $f(\mathbb{R})$, which shows that the equation $g(f(x)) = 0$ has a real root, a contradiction. This proves the claim. $\square$

If $\alpha_2 \leq \beta_2$, then by the Claim, the equation $f(f(x)) = 0$ has no real roots. Similarly, if $\beta_2 \leq \alpha_2$, then the equation $g(g(x)) = 0$ has no real roots. This completes the proof. $\blacksquare$

Example 11.5. Any polynomial of odd degree with real coefficients has a real root.

Example 11.6. Let $P(x)$ and $Q(x)$ be monic polynomials of degree 10 having real coefficients. Assume that the equation $P(x) = Q(x)$ has no real roots. Prove that the equation $P(x+1) = Q(x-1)$ has at least one real root.

Walkthrough —

- (a) Consider the difference $P(x) - Q(x)$ to show that the coefficient of x^9 in these polynomials are equal.
- (b) Prove that the polynomial $P(x+1) - Q(x-1)$ is of degree 9.

Solution 68. Note that $P(x) - Q(x)$ is a polynomial of degree at most 9 having real coefficients. Since $P(x) - Q(x)$ has no real root, it follows that it has degree at most 8. In other words, the coefficients of x^9 in $P(x), Q(x)$ are the same. Note that $P(x+1) - Q(x-1)$ is of degree ≤ 9 , and the coefficient of

x^9 in $P(x+1) - Q(x-1)$ is equal to the coefficient of x^9 in $(x+1)^{10} - (x-1)^{10}$, which is equal to 20. This shows that $P(x+1) - Q(x-1)$ is a polynomial of degree 9 with real coefficients. Consequently, it has at least one real root. ■

Example 11.7. Let $P(x)$ be a nonconstant polynomial with real coefficients having a real root. Suppose it does not vanish at 0. Show that the monomial terms appearing in $P(x)$ can be erased one by one to obtain its constant term such that the intermediate polynomials have at least one real root.

Walkthrough —

(a)

Solution 69. Write $P(x) = a_n x^n + \dots + a_0$ with $a_n, \dots, a_0$ lying in $\mathbb{R}$, and $a_n a_0 \neq 0$. Since any polynomial of odd degree has a real root, it follows that if the degree of $P(x)$ is odd, then the nonconstant monomials, other than the leading term, can be erased one by one, and then, the leading term can be erased to obtain a_0 , and the intermediate polynomials have a real root. If a_n, a_0 are of opposite signs and n is even, then the same process can be followed, and note that any of the intermediate polynomials takes values of opposite signs at 0 and at a large enough integer, and hence has a real root.

Let us assume that n is even, and a_n, a_0 are of the same sign. Let α denote a real root of $P(x)$. Let $Q(x)$ denote the polynomial $P(x) - a_n x^n$. Note that $Q(0) = a_0$ and $Q(\alpha) = -a_n \alpha^n$. Since a_0 is nonzero, it follows that $\alpha \neq 0$, and hence, $Q(0), Q(\alpha)$ are of opposite signs. This shows that $Q(x)$ is a nonconstant polynomial with real coefficients having a real root, and it does not vanish at 0. Since the degree of $Q(x)$ is smaller than that of $P(x)$, by induction, we are done. ■

Exercise 11.8 (China TST 1995 Day 2 P2, AoPS). Alice and Bob play a game with a polynomial of degree at least 4:

$$x^{2n} + \square x^{2n-1} + \square x^{2n-2} + \dots + \square x + 1.$$

They take turns to fill the empty boxes. If the resulting polynomial has no real root, Alice wins, otherwise, Bob wins. If Alice goes first, who has a winning strategy?

Walkthrough —

- (a) There are more odds than evens among the integers $1, 2, \dots, 2n-1$.
- (b) Can Bob have a winning strategy using the odds in his favour?

Solution 70. Bob has a winning strategy, as described below.

Bob makes sure that at the end of each of his turns except the last one, the number of even powers of x whose coefficients have been provided by some of them is equal to the number of odd powers of x whose coefficients have been provided by some of them. This can be done, for instance, if during a turn of Bob, other than the last turn, Bob provides the coefficient of an odd (resp. even) power of x if Alice has provided the coefficient of an even (resp. odd) power of x in the preceding turn.

Since $n \geq 2$, it follows that Bob gets at least one turn. At the beginning of the final turn of Bob, there are two powers of x whose coefficients are to be determined, denote them by x^i, x^j , their coefficients by c_i, c_j respectively. Let $Q(x)$ denote the polynomial, obtained by the erasing the terms corresponding to x^i, x^j from the polynomial that Bob had at the beginning of his final turn. Note that

$$P(x) = Q(x) + c_i x^i + c_j x^j.$$

Note that at least of i, j is odd. Interchanging i, j if necessary, let us assume that i is odd. We describe the strategy that Bob follows in the two cases below.

Let us consider the case that j is even. Bob determines c_j in such a way that for any choice of c_i , the completed polynomial $P(x)$ is guaranteed to have at least one real root. This can be done, for instance, by taking c_j satisfying

$$Q(1) + Q(-1) + 2c_j = 0.$$

For any choice of c_i , the above choice of c_j shows that $P(1) + P(-1) = 0$, which implies that $P(x)$ has a root in the interval $[-1, 1]$.

Let us consider the case that j is odd. Bob determines c_j in such a way that for any choice of c_i by Alice in the next turn, the completed polynomial $P(x)$ is guaranteed to have at least one real root. This can be done, for instance, by taking c_j satisfying

$$Q(2) + c_j 2^j + 2^i Q(-1) - c_j 2^i = 0.$$

Since $i \neq j$, the above holds for some $c_j \in \mathbb{R}$. For any choice of c_i , note that

$$P(2) + 2^i P(-1) = 0$$

holds, which implies that $P(x)$ has a root in $[-1, 2]$. ■

Exercise 11.9 (Bay Area MO 2004 P5, AoPS). Find (with proof) all monic polynomials $f(x)$ with integer coefficients that satisfy the following two conditions.

1. $f(0) = 2004$.
2. If x is irrational, then $f(x)$ is also irrational.

Solution 71. Let $f(x)$ be a polynomial satisfying the required conditions. Note that $f(x)$ is nonconstant. If $f(x)$ is linear, then it is equal to $x + 2004$. It remains to consider the case when the degree of $f(x)$ is at least 2, which we assume from now on.

Let n be a positive integer and p be a prime number satisfying

$$f(0) - n = -p.$$

Since $f(x) - n$ takes positive values for large positive x and a negative value at $x = 0$, by the intermediate value theorem, the polynomial $f(x) - n$ has a real root α . By the given conditions, α is rational. Since $f(x) - n$ is a monic polynomial with integer coefficients and its constant term is $-p$, by the rational root theorem, α is an integer and it divides p . Since p is prime, it follows that α is either 1 or -1 or p or $-p$. Write

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + f(0).$$

Note that for any integer k ,

$$\begin{aligned} |f(k) - n| &\geq |f(0) - n| - |k^n + a_{n-1}k^{n-1} + \cdots + a_1k| \\ &\geq p - (k^n + |a_{n-1}|k^{n-1} + \cdots + |a_1|k) \\ &\geq p - (1 + |a_{n-1}| + \cdots + |a_1|)k^n, \end{aligned}$$

and

$$\begin{aligned} |f(k) - n| &\geq |k^n + a_{n-1}k^{n-1} + \cdots + a_1k| - |f(0) - n| \\ &\geq |k^n| - (|a_{n-1}||k^{n-1}| + \cdots + |a_1||k|) - p \\ &\geq |k^n| - (|a_{n-1}| + \cdots + |a_1|)|k^{n-1}| - p \end{aligned}$$

hold.

If p is large enough, then so is n , and hence $f(x) - n$ does not vanish at any of $1, -1, p, -p$. Indeed, if

$$p > 1 + |a_{n-1}| + \cdots + |a_1|,$$

then $f(1) - n$ and $f(-1) - n$ are nonzero, and $f(p) - n$ and $f(-p) - n$ have absolute values at least

$$p^{n-1}(p - (|a_{n-1}| + \cdots + |a_1|)) - p > p^{n-1} - p \geq 0,$$

where the last inequality holds since $n \geq 2$. This contradicts the fact that α is one of $1, -1, p, -p$. Therefore, no polynomial $f(x)$ of degree at least 2 satisfies the required conditions. This proves that any polynomial $f(x)$ satisfying the required conditions is linear, and is equal to $x + 2004$.

Also note that the polynomial $x + 2004$ indeed satisfies the required conditions.

Hence, the only polynomial $f(x)$ satisfying the required conditions is

$$f(x) = x + 2004.$$

■

§12 Lagrange interpolation

Lemma 6

Let $x_1, \dots, x_n$ be pairwise distinct real numbers, and $y_1, \dots, y_n$ be real numbers. Then there exists a unique polynomial $P(x)$ of **degree at most $n - 1$** having real coefficients such that $P(x_i) = y_i$ for all $1 \leq i \leq n$. Moreover, this statement also holds if the reals are replaced by rationals or complex numbers all throughout.

Proof. Note that there is at most one polynomial satisfying the required condition. Observe that the polynomial $P(x)$, defined by

$$P(x) = \sum_{i=1}^n y_i \prod_{j \neq i} \frac{x - x_j}{x_i - x_j},$$

satisfies the required condition. □

Example 12.1. If a polynomial of degree n takes rationals to rationals on $n + 1$ points, then show that it is a rational polynomial.

Walkthrough —

(a) Apply the Lagrange interpolation formula.

Solution 72. ■

Lemma 7

Let $x_1, \dots, x_n$ be pairwise distinct real numbers, and $y_1, \dots, y_n$ be real numbers. Then there exists a unique **monic** polynomial $P(x)$ of **degree n** having real coefficients such that $P(x_i) = y_i$ for all $1 \leq i \leq n$. Moreover, this statement also holds if the reals are replaced by rationals or complex numbers all throughout.

Proof. By the above lemma, there exists a polynomial $Q(x)$ of degree at most $n - 1$ with real coefficients such that $Q(x_i) = y_i - x_i^n$ for all $1 \leq i \leq n$. Write $P(x) = x^n + Q(x)$. Note that $P(x)$ is a monic polynomial of degree n with real coefficients and $P(x_i) = y_i$ for all $1 \leq i \leq n$. □

Example 12.2. Suppose $P(x)$ is a monic polynomial of degree $n - 1$ with real coefficients. Let $a_1, a_2, \dots, a_n$ be distinct real numbers. Show that

$$\sum_{i=1}^n \frac{P(a_i)}{\prod_{j \neq i} (a_j - a_i)} = 1.$$

Solution 73. For $1 \leq i \leq n$, write $y_i = P(a_i)$. Note that

$$P(x) = \sum_{i=1}^n y_i \prod_{j \neq i} \frac{x - a_i}{a_i - a_j}.$$

Comparing the leading coefficients, the result follows. ■

Example 12.3. Let $P(x)$ be a monic polynomial of degree n . Show that

$$\sum_{i=0}^n (-1)^{n-i} \binom{n}{i} P(i) = n!.$$

Walkthrough — Is the above of some use?

Exercise 12.4 (USAMO 2002 P3, AoPS). Prove that any monic polynomial (a polynomial with leading coefficient 1) of degree n with real coefficients is the average of two monic polynomials of degree n with n real roots.

Walkthrough —

- (a) Let $F(x)$ be a monic polynomial of degree n with real coefficients. We would like to write

$$2F(x) = P(x) + Q(x),$$

where $P(x), Q(x)$ are polynomials with certain properties.

- (b) Let us take $P(x)$ to be a polynomial which changes sign very often, so that it is likely to have n real roots. To do so, choose n real numbers satisfying

$$x_1 < x_2 < \cdots < x_n,$$

and let $y_1, \dots, y_n$ be real numbers (to be specified later). Apply the Lagrange interpolation formula to obtain a monic polynomial $P(x)$ satisfying $P(x_i) = y_i$ for all i .

- (c) Define the polynomial $Q(x)$ using

$$2F(x) = P(x) + Q(x).$$

Note that $Q(x)$ is a monic polynomial with real coefficients.

- (d) Can one impose suitable conditions on $y_1, \dots, y_n$ such that $Q(x)$ changes sign often?

Solution 74. Let $F(x)$ be a monic polynomial of degree n with real coefficients. Let $x_1 < \cdots < x_n$ be real numbers. Note that there exist real numbers $y_1, \dots, y_n$ satisfying

$$(-1)^i y_i > 0, \quad (-1)^{i-1} (2F(x_i) - y_i) > 0$$

for any $1 \leq i \leq n$. Indeed, y_i 's can be taken satisfying $(-1)^i y_i > |F(x_i)|$. Let $P(x)$ be a monic polynomial of degree n with real coefficients such that $P(x_i) = y_i$ for all $1 \leq i \leq n$. Let $Q(x)$ denote the polynomial such that $F(x)$ is the average of $P(x)$ and $Q(x)$. Since $F(x), P(x)$ are monic, it follows that so is $Q(x)$. For any $1 \leq i \leq n$, note that $Q(x_i) = 2F(x_i) - P(x_i) = 2F(x_i) - y_i$ holds, and hence $Q(x_i)$ has sign as that of $(-1)^i$. It follows that each of the polynomials $P(x), Q(x)$ has at least one root in each of the intervals

$$(x_1, x_2), (x_2, x_3), \dots, (x_{n-1}, x_n).$$

Since these polynomials are of degree n with real coefficients and each of them has at least $n - 1$ real roots, all of their roots are real. ■

Exercise 12.5 (Putnam 1968 A6, AoPS). Find all polynomials whose coefficients are all ± 1 and whose roots are all real.

Walkthrough —

- (a) Consider the average of the squares of the roots, and show that it is small (and consequently, smaller than their geometric mean) if the polynomial has degree ≥ 4 .
- (b) Repeat the argument for degree three polynomials.
- (c) Finding the degree one and degree two polynomials is easy.

Solution 75. Let $P(x)$ be a polynomial of degree n having real roots. Assume that its coefficients are equal to ± 1 . Denote its roots by $\alpha_1, \dots, \alpha_n$, counting multiplicities. Noting

$$\alpha_1^2 + \dots + \alpha_n^2 = (\alpha_1 + \dots + \alpha_n)^2 - 2 \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j,$$

it follows that

$$\alpha_1^2 + \dots + \alpha_n^2 = 3.$$

Applying the AM-GM inequality, we obtain

$$\alpha_1^2 + \dots + \alpha_n^2 \geq n,$$

which implies that $n \leq 3$.

If $P(x)$ is a monic linear polynomial, then it is equal to one of $x - 1, x + 1$.

If $P(x)$ is a monic quadratic polynomial, then considering discriminants, it follows that its constant term is equal to 1, and hence $P(x)$ is equal to one of $x^2 + x - 1, x^2 - x - 1$.

Let us consider the case that $P(x)$ is a monic cubic polynomial, and the coefficient of x^2 in $P(x)$ is equal to 1. Note that $x^3 + x^2 + x - 1$ vanishes at $x = 0$. For any real number $\alpha \leq 0$,

$$\alpha^3 + \alpha^2 + \alpha - 1 = \alpha(\alpha + 1)^2 - 1 - \alpha^2 \leq -1$$

holds, which implies that any real root of $x^3 + x^2 + x - 1$ is positive. Since $x^3 + x^2 + x - 1$ is of odd degree, it has a real root, and since $x \mapsto x^3 + x^2 + x - 1$ is an increasing function on the set of positive reals, it follows that $x^3 + x^2 + x - 1$ has only one real root. This gives $P(x) \neq x^3 + x^2 + x - 1$.

Let us consider the case that $P(x) = x^3 + x^2 - x + 1$ Using

$$P(x) = (x^2 - x) + 1 + x^3,$$

$$P(x) = x^3 + x^2 + (1 - x),$$

$$P(x) = x^3 + (x^2 - x) + 1,$$

it follows that $P(x)$ has no root in $[-1, \infty)$. Note that for $-1 \geq a \geq b$, we have

$$\begin{aligned} P(a) - P(b) &= (a - b)(a^2 + ab + b^2 + a + b - 1) \\ &= (a - b)((a^2 + a) + (b^2 + b) + (ab - 1)) \\ &\geq 0. \end{aligned}$$

This shows that $x^3 + x^2 - x + 1$ has no real root, and hence $P(x) \neq x^3 + x^2 - x + 1$.

Note that not all the roots of

$$x^3 + x^2 + x + 1 = (x + 1)(x^2 + 1)$$

are real. Consequently, if $P(x)$ is a monic cubic polynomial and the coefficient of x^2 in $P(x)$ is equal to 1, then $P(x)$ is equal to $x^3 + x^2 - x - 1$.

If $P(x)$ is a monic cubic polynomial and the coefficient of x^2 in $P(x)$ is -1 , then $-P(-x)$ is equal to $x^3 + x^2 - x - 1$, and hence $P(x)$ is equal to $x^3 - x^2 - x + 1$.

This shows that $P(x)$ is equal to one of

$$x - 1, x + 1, x^2 + x - 1, x^2 - x - 1, x^3 + x^2 - x - 1, x^3 - x^2 - x + 1.$$

Note that the discriminants of the quadratic polynomial $x^2 + x - 1, x^2 - x - 1$ are nonnegative, and also note that

$$\begin{aligned} x^3 + x^2 - x - 1 &= (x - 1)(x + 1)^2, \\ x^3 - x^2 - x + 1 &= (x - 1)^2(x + 1). \end{aligned}$$

Consequently, the roots of the polynomials

$$x - 1, x + 1, x^2 + x - 1, x^2 - x - 1, x^3 + x^2 - x - 1, x^3 - x^2 - x + 1$$

are all real. Hence, the required polynomials are

$$\begin{aligned} &x - 1, x + 1, x^2 + x - 1, x^2 - x - 1, x^3 + x^2 - x - 1, x^3 - x^2 - x + 1, \\ &-(x - 1), -(x + 1), -(x^2 + x - 1), -(x^2 - x - 1), -(x^3 + x^2 - x - 1), -(x^3 - x^2 - x + 1). \end{aligned}$$

■

§13 Integer divisibility

Example 13.1. If P is a polynomial with integer coefficients and a, b are integers, then $P(a) - P(b)$ is a multiple of $a - b$.

Walkthrough —

- (a) Show that it suffices to prove it for monomials. ■

Solution 76.

Example 13.2. Let $P(x)$ be a polynomial with integer coefficients such that $P(0), P(1)$ are odd. Show that $P(x)$ does not have any integer root.

Solution 77. If $P(x)$ admits an integer root α , then α divides $P(0)$ and $\alpha - 1$ divides $P(1)$, which shows that $\alpha, \alpha - 1$ are odd, which is impossible. ■

Exercise 13.3 (IOQM 2023 P12, AoPS). Let $P(x) = x^3 + ax^2 + bx + c$ be a polynomial where a, b, c are integers and c is odd. Let p_i be the value of $P(x)$ at $x = i$. Given that $p_1^3 + p_2^3 + p_3^3 = 3p_1p_2p_3$, find the value of $p_2 + 2p_1 - 3p_0$.

Summary — Using parity conditions, show that $p_1 + p_2 + p_3 \neq 0$. This gives $p_1 = p_2 = p_3$. This gives a system of equations in a and b , which can be solved to find a and b .

Walkthrough —

- (a) Note that if $p_1 + p_2 + p_3 \neq 0$, then using the condition that p_1, p_2, p_3 are real, it follows that $p_1 = p_2 = p_3$.
 (b) Note that

$$\begin{aligned} p_1 &= 1 + a + b + c, \\ p_2 &= 8 + 4a + 2b + c, \\ p_3 &= 27 + 9a + 3b + c, \end{aligned}$$

which gives that

$$p_1 + p_2 + p_3 = 36 + 14a + 6b + 3c.$$

Since c is an odd integer, it follows that ^a $p_1 + p_2 + p_3 \neq 0$. Consequently, $p_1 = p_2 = p_3$.

- (c) This gives

$$3a + b = -7, 5a + b = -19,$$

and hence, $a = -6, b = 11$.

(d) This gives

$$\begin{aligned} p_2 + 2p_1 - 3p_0 &= 10 + 6a + 4b + 3c - 3c \\ &= 18. \end{aligned}$$

^aSince c is odd, it follows that p_2 is odd. Note that $p_1 - p_3$ is even (Why? Does it follow that $f(1) - f(3)$ is divisible by 2 for any polynomial $f(x)$ with integer coefficients?), and hence, $p_1 + p_2 + p_3$ is odd.

Solution 78. Since $P(x)$ is a polynomial with integer coefficients, it follows that $p_1 - p_3$ is divisible by 2. Since c is odd, it follows that p_2 is odd. This shows that $p_1 + p_2 + p_3$ is odd, and hence, $p_1 + p_2 + p_3 \neq 0$. Using $p_1^3 + p_2^3 + p_3^3 = 3p_1p_2p_3$, we obtain

$$(p_1 + p_2 + p_3)(p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1) = 0.$$

This gives

$$p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1 = 0,$$

or equivalently,

$$(p_1 - p_2)^2 + (p_2 - p_3)^2 + (p_3 - p_1)^2 = 0,$$

which implies that $p_1 = p_2 = p_3$. This shows that

$$1 + a + b + c = 8 + 4a + 2b + c = 27 + 9a + 3b + c,$$

which yields

$$3a + b = -7, 5a + b = -19.$$

This gives $a = -6, b = 11$. It follows that

$$\begin{aligned} p_2 + 2p_1 - 3p_0 &= 10 + 6a + 4b + 3c - 3c \\ &= 10 - 36 + 44 \\ &= 18. \end{aligned}$$

■

Example 13.4 (India RMO 2016g P8). At some integer points a polynomial with integer coefficients take values 1, 2 and 3. Prove that there exist not more than one integer at which the polynomial is equal to 5.

Solution 79. Denote the polynomial by $P(x)$. On the contrary, let us assume that there are at least two distinct integers where $P(x)$ takes the value 5.

Let a, b, c be integers such that

$$P(a) = 1, P(b) = 2, P(c) = 3.$$

Note that $a - b$ divides $P(a) - P(b)$, $b - c$ divides $P(b) - P(c)$. It follows that $a - b = \pm 1$, $b - c = \pm 1$. Since a, b are of opposite parity, and so are the integers b, c , we obtain that a, c are of the same parity. Noting that $c - a$ divides $P(c) - P(a) = 2$, it follows that $c - a = \pm 2$. Combining this with $a - b = \pm 1$, $b - c = \pm 1$, we get $a - b = b - c = 1$ or $a - b = b - c = -1$.

This shows that $P(b - 1) = 1$, $P(b) = 2$, $P(b + 1) = 3$ holds or $P(b + 1) = 1$, $P(b) = 2$, $P(b - 1) = 3$ holds. Note that in the first case, the polynomial $R(x) := P(x - b)$ takes the values 1, 2, 3 at the integers $-1, 0, 1$ respectively. In the second case, the polynomial $S(x) = P(-x + b)$ takes the values 1, 2, 3 at the integers $-1, 0, 1$ respectively. This proves that there is a polynomial $Q(x)$ with integer coefficients which takes the values 1, 2, 3 at $-1, 0, 1$ respectively.

From the hypothesis, it follows that there are distinct integers i, j such that $Q(i) = Q(j) = 5$. Note that $i - 1$ divides $Q(i) - Q(1) = 2$, i divides $Q(i) - Q(0) = 3$, $i + 1$ divides $Q(i) - Q(-1) = 4$. Since i divides 3, we obtain $i = \pm 1, \pm 3$. Using $Q(-1) = 1$, $Q(1) = 3$, we get $i \neq -1$, $i \neq 1$. This gives $i = \pm 3$. Noting that $i - 1$ divides 2, we obtain $i \neq -3$, and hence $i = 3$. Similarly, it follows that $j = 3$. ■

Example 13.5. Let $P(x)$ be a polynomial with integer coefficients such that $P(20), P(25)$ are of absolute value equal to 1. Show that $P(x)$ does not vanish at any integer.

Walkthrough —

- (a) If $P(x)$ vanishes at an integer α , then $\alpha - 20$ divides $P(20)$ and $\alpha - 25$ divides $P(25)$.

Solution 80. On the contrary, let us assume that $P(x)$ vanishes at an integer α . Note that $\alpha - 20$ divides 1, and so does $\alpha - 25$. This shows that $\alpha - 20, \alpha - 25$ are of absolute value equal to 1. Applying triangle inequality, we obtain

$$5 \leq |\alpha - 20| + |\alpha - 25| \leq 2,$$

which is impossible. ■

Exercise 13.6 (USAMO 1974 P1, AoPS). Let a, b , and c denote three distinct integers, and let P denote a polynomial having all integral coefficients. Show that it is impossible that $P(a) = b$, $P(b) = c$, and $P(c) = a$.

Walkthrough —

- (a) Show that each of the integers $a - b, b - c, c - a$ is a multiple of the remaining two integers.
 (b) Prove that this implies that a, b, c are equal.

Solution 81. Note that

$$a - b \mid P(a) - P(b) = b - c \mid P(b) - P(c) \mid c - a \mid P(c) - P(a) = a - b.$$

Consequently, the integers $a - b, b - c, c - a$ are of the same absolute value. Denote their absolute value by k . Note that their sum is zero. However, the sum is equal to mk , for some $m \in \{\pm 1, \pm 3\}$. Hence, k is equal to zero. This yields that $a = b = c$. ■

Here is a more general result.

Example 13.7. Let $P(x)$ be a polynomial with integer coefficients, and let n be an odd positive integer. Suppose that $x_1, x_2, \dots, x_n$ is a sequence of integers such that $x_2 = P(x_1), x_3 = P(x_2), \dots, x_n = P(x_{n-1})$, and $x_1 = P(x_n)$. Prove that all the x_i 's are equal.

Walkthrough — Show that

$$a_1 - a_2 \mid a_2 - a_3 \mid a_3 - a_4 \mid \dots \mid a_n - a_1 \mid a_1 - a_2.$$

Note that sum of these differences is an odd multiple of their absolute value.

Exercise 13.8 (Tournament of Towns Spring 2014, Senior A Level P4, by G.K. Zhukov). In the plane, the points with integer coordinates (x, y) satisfying $0 \leq y \leq 10$ are marked. Consider a polynomial of degree 20 with integer coefficients. Determine the maximum possible number of marked points which can lie on its graph.

Walkthrough —

(a)

Solution 82. Note that the polynomial

$$(x - 1)(x - 2)(x - 3) \dots (x - 20)$$

of degree 20 has integer roots. Let us prove that the graph of no polynomial of degree 20 with integer coefficients passes through more than 20 marked points.

Claim — Let $P(x)$ be a polynomial of degree 20 with integer coefficients. No more than 20 marked points lie on the graph of $P(x)$.

Proof of the Claim. On the contrary, let us assume that there are integers $x_1 < x_2 < \dots < x_{21}$ such that

$$0 \leq P(x_i) \leq 10$$

holds for all $1 \leq i \leq 21$. For any integer $1 \leq i \leq 10$, the inequality $x_{21} - x_i \geq 11$ holds, and using that $x_{21} - x_i$ divides the integer $P(x_{21}) - P(x_i)$, which lies in $[-10, 10]$, it follows that $P(x_{21}) = P(x_i)$. Similarly, for any integer $12 \leq i \leq 21$, it follows that $P(x_1) = P(x_i)$. This shows that $P(x_i) = P(x_1)$ for any integer $i \in \{1, 2, \dots, 10\} \cup \{12, 13, \dots, 21\}$. Since $P(x)$ is a polynomial of degree 20, it follows that

$$P(x) - P(x_1) = c(x - x_1)(x - x_2) \dots (x - x_{10})(x - x_{12})(x - x_{13}) \dots (x - x_{21})$$

holds for some nonzero integer a . This yields that

$$|P(x_{11}) - P(x_1)| \geq (10!)^2,$$

which is impossible. □

This proves that the maximum possible number of marked points which can lie on the graph of a polynomial of degree 20 with integer coefficients is equal to 20. ■

Exercise 13.9 (RMO 2025a P6, AoPS). Let $p(x)$ be a nonconstant polynomial with integer coefficients, and let $n \geq 2$ be an integer such that no term of the sequence

$$p(0), p(p(0)), p(p(p(0))), \dots$$

is divisible by n . Show that there exist integers a, b such that $0 \leq a < b \leq n - 1$ and n divides $p(b) - p(a)$.

Walkthrough —

- (a) Apply the pigeonhole principle to show that some two terms of the given sequence are congruent modulo n .
- (b) Let i and j be positive integers such that $i < j$ and the i -th and j -th terms of the sequence are congruent modulo n .
- (c) Reduce the arguments of these two terms modulo n to obtain integers a and b such that $0 \leq a, b \leq n - 1$ and n divides $p(b) - p(a)$.
- (d) Show that $a \neq b$. It could be useful to assume that the integer i as above is the smallest with this property.

Solution 83. Define the sequence

$$p^{(0)}(0), p^{(1)}(0), p^{(2)}(0), p^{(3)}(0), \dots$$

by

$$p^{(n)}(0) := \begin{cases} 0 & \text{if } n = 0, \\ p(p^{(n-1)}(0)) & \text{if } n \geq 1. \end{cases}$$

By the pigeonhole principle, some two among the following $n + 1$ integers

$$p^{(1)}(0), p^{(2)}(0), \dots, p^{(n+1)}(0)$$

are congruent modulo n . Let i denote the smallest positive integer such that there exists a positive integer $j > i$ with

$$p^{(i)}(0) \equiv p^{(j)}(0) \pmod{n}.$$

Let $0 \leq a \leq n - 1$ and $0 \leq b \leq n - 1$ be integers such that

$$a \equiv p^{(i-1)}(0) \pmod{n}, \quad b \equiv p^{(j-1)}(0) \pmod{n}.$$

Note that n divides $p(b) - p(a)$. We claim that $a \neq b$. Suppose, for the sake of contradiction, that $a = b$. It follows that

$$p^{(i-1)}(0) \equiv p^{(j-1)}(0) \pmod{n}.$$

Using $j - 1 > i - 1$ and the minimality of i , we obtain $i = 1$. This implies that

$$p^{(0)}(0) \equiv p^{(j-1)}(0) \pmod{n}.$$

Since $p^{(0)}(0) = 0$ and $j - 1 > 0$, we have that n divides $p^{(j-1)}(0)$, contradicting the given condition. This implies that $a \neq b$.

Reordering if necessary, we may assume that $a < b$. Consequently, there exist integers $0 \leq a < b \leq n - 1$ such that n divides $p(b) - p(a)$. ■

Lemma 8

Let P be a polynomial with integer coefficients. Suppose a is an integer and k is a positive integer such that $P^k(a) = a$, where P^k denotes the k -fold composite map from $\mathbb{Z} \rightarrow \mathbb{Z}$. Show that $P^2(a) = a$.

Proof. Let ℓ denote the smallest positive integer such that $P^\ell(a) = a$. If $\ell = 1$ or $\ell = 2$, then we are done. Henceforth, we assume that $\ell \geq 3$.

Note that

$$P(a) - a \mid P^2(a) - P(a) \mid \dots \mid P^\ell(a) - P^{\ell-1}(a) = a - P^{\ell-1}(a) \mid P(a) - a.$$

Since $a - P^{\ell-1}(a)$ is nonzero, it follows that the above differences are nonzero. Consequently, for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = \pm(P^i(a) - P^{i-1}(a)).$$

If $P^{i+1}(a) = P^{i-1}(a)$ holds for some $1 \leq i \leq \ell$, then applying $P^{\ell-i+1}$ to both sides, we obtain $P^2(a) = a$, which contradicts the assumption that $\ell \geq 3$. It follows that for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = P(a) - a$$

holds, which implies that

$$\sum_{i=0}^{\ell-1} (P^{i+1}(a) - P^i(a)) = \ell(P(a) - a).$$

This gives $P(a) = a$, which contradicts the assumption that $\ell \geq 3$. This completes the proof. $\square$

Exercise 13.10 (IMO 2006 P5, AoPS, by Dan Schwarz, Romania). Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients, and let k be a positive integer. Consider the polynomial $Q(x) = P(P(\dots P(P(x)) \dots))$, where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.

Walkthrough —

(a) Does the above lemma help?

Solution 84. By the above lemma, it reduces to considering the case $Q(x) = P^2(x)$.

Suppose Q has more than n fixed points. Since P is not linear, it follows that P cannot have n fixed points, and hence not all the fixed points of Q are fixed points of P . Let b be a non-fixed point of P , and $Q(b) = b$. Suppose a be a fixed point of Q , other than b .

Let us first consider the case that $P(a) \neq a$. Note that

$$P(b) - a \mid P(a) - b \mid P(b) - a$$

holds, and

$$a - b \mid P(a) - P(b) \mid a - b$$

holds too. This yields that

$$|P(a) - b| = |P(b) - a|, \quad |P(a) - P(b)| = |a - b|.$$

If

$$P(a) - b = a - P(b), \text{ and } P(a) - P(b) = a - b$$

hold, then b would be a fixed point of P . It follows that at least one of

$$P(a) - b = -(a - P(b)), P(a) - P(b) = -(a - b)$$

holds. Consequently, we obtain

$$P(a) + a = P(b) + b.$$

Next, let us consider the case that $P(a) = a$. Note that

$$P(b) - a \mid b - a \mid P(b) - a.$$

Since b is not a fixed point for P , it follows that

$$P(b) - a = a - b,$$

which yields

$$P(a) + a = P(b) + b.$$

This proves that all the roots of $Q(x) = x$ are the roots of $P(x) + x = P(b) + b$. Since $P(x)$ has degree $n > 1$, it follows that the polynomial $P(x) + x - P(b) - b$ is of degree n , and it has more than n roots, which is impossible.

Hence, there are at most n integers t such that $Q(t) = t$ holds. ■

§14 Functional equations

Example 14.1 (cf. [Canadian Mathematical Olympiad 1969 P8](#), [Australian Mathematics Competition 1984](#), [India Pre-RMO 2012 P16](#)). [[Tao06](#), Problem 3.2] Suppose

$$x_1, x_2, x_3, x_4, \dots$$

is a sequence of integers satisfying the following properties:

- (1) $x_2 = 2$,
- (2) $x_{mn} = x_m x_n$ for all positive integers m, n ,
- (3) $x_m < x_n$ for any positive integers m, n with $m < n$.

Find x_{2024} . What is the value of the sum

$$x_1 + x_2 + \dots + x_{2024}?$$

Summary — Observe that $x_{2^n} = 2^n$ for any $n \geq 1$. Combining this with the hypothesis that $\{x_n\}_{n \geq 1}$ is an increasing sequence of **positive integers**, conclude that $x_n = n$ for any $n \geq 1$.

Walkthrough —

- (a) What can be said about x_4, x_8, x_{16}, x_{32} ?
- (b) Note that $x_4 = x_{2 \times 2}, x_8 = x_{4 \times 2}, x_{16} = x_{8 \times 2}, x_{32} = x_{16 \times 2}$.
- (c) Can one show that $x_{2^n} = 2^n$ for any $n \geq 1$?
- (d) Show that $x_m = m$ for any $m \geq 1$ (does property (3) help?).

Solution 85. From the second condition, we obtain

$$x_{2^n} = x_2^n$$

for any integer $n \geq 1$. Using the first condition, it follows by induction that

$$x_{2^n} = 2^n$$

for any integer $n \geq 1$. Since $\{x_n\}_{n \geq 1}$ is an increasing sequence of positive integers, it follows that $x_n = n$ for any positive integer n . This gives

$$x_{2024} = 2024.$$

It follows that

$$x_1 + x_2 + \cdots + x_{2024} = 1012 \cdot 2025 = 2049300.$$

■

Exercise 14.2 (Putnam 1963 A2, AoPS). Let $f: \mathbb{N} \rightarrow \mathbb{N}$ be a strictly increasing function such that $f(2) = 2$ and $f(mn) = f(m)f(n)$ for every relatively prime pair of positive integers m and n . Show that $f(n) = n$ for every positive integer n .

Walkthrough —

(a)

Solution 86. Note that $f(1) = 1$ since $f(2) = f(1)f(2)$. Write $f(3) = k$. Note that

$$\begin{aligned} k(k+2) &\leq f(3)f(5) \\ &= f(15) \\ &\leq f(18) - 3 \\ &= f(2)f(9) - 3 \\ &= 2(f(10) - 1) - 3 \\ &= 2f(2)f(5) - 5 \\ &= 4f(5) - 5 \\ &\leq 4(f(6) - 1) - 5 \\ &= 4f(2)f(3) - 9 \\ &= 8k - 9 \end{aligned}$$

holds, which implies that $k^2 - 6k + 9 \leq 0$, which gives $k = 3$.

Claim — For any integer $n \geq 3$, we have $f(n) = n$.

Proof of the Claim. Note that $f(n) = n$ holds for $n = 3$. Assume that $m \geq 3$ is an integer such that $f(n) = n$ holds for all $1 \leq n \leq m$.

Let us first consider the case that m is odd. Then,

$$f(2m) = 2f(m) = 2m$$

holds. Since f is strictly increasing, using $f(m) = m$ and $f(2m) = 2m$, it follows that

$$f(m+1) = m+1.$$

Next, consider the case that m is even. Then, it follows that $m-1 \geq 3$ and $f(m-1) = m-1$. This gives

$$f(2(m-1)) = 2f(m-1) = 2(m-1).$$

Since f is strictly increasing, using $f(m-1) = m-1$ and $f(2(m-1)) = 2(m-1)$, and noting that $m < 2(m-1)$, it follows that

$$f(m) = m.$$

By the principle of induction, the claim holds. □

■

Using $f(3) = 3$ and that

$$f(2^{n+1} + 2) = 2f(2^n + 1)$$

holds for any integer $n \geq 1$, one can also prove by induction that $f(2^n + 1) = 2^n + 1$ for any integer $n \geq 1$. The monotonicity of f then implies that $f(n) = n$ for all positive integers n .

Example 14.3 (India RMO 2006 P7). Let X be the set of all positive integers greater than or equal to 8, and let $f: X \rightarrow X$ be a function such that $f(x+y) = f(xy)$ for all $x \geq 4, y \geq 4$. If $f(8) = 9$, determine $f(9)$.

Walkthrough — Using the given condition, try to express $f(9)$ in terms of $f(8)$.

Solution 87. Note that

$$f(9) = f(4+5) = f(20) = f(4+16) = f(64),$$

and

$$f(8) = f(4+4) = f(16) = f(8+8) = f(64)$$

hold. Using the given condition $f(8) = 9$, it follows that $f(9) = 9$. ■

§14.1 Using symmetry

Example 14.4. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$x^2 f(x) + f(1-x) = 2x - x^4$$

holds for all $x \in \mathbb{R}$.

Solution 88. Let f be a function satisfying the given functional equation. Let x be an element of $\mathbb{R}$. Note that

$$\begin{aligned} x^2 f(x) + f(1-x) &= 2x - x^4, \\ f(x) + (1-x)^2 f(1-x) &= 2(1-x) - (1-x)^4 \end{aligned}$$

holds. Eliminating $f(1-x)$ from these two equations, we get

$$\begin{aligned} ((x(1-x))^2 - 1)f(x) &= (2x - x^4)(1-x)^2 - 2(1-x) + (1-x)^4 \\ &= (1-x)((1-x)^3 + (2x - x^4)(1-x) - 2) \\ &= (1-x)(1 - 3x + 3x^2 - x^3 + 2x - 2x^2 - x^4 + x^5 - 2) \\ &= (1-x)(-1 - x + x^2 - x^3 - x^4 + x^5) \\ &= (1-x)(x+1)(-1 - x^3 + x^2(x^2 - x + 1)) \\ &= (1-x^2)(-1 - x^3 + x^4 - x^3 + x^2) \\ &= (1-x^2)(x^4 - 2x^3 + x^2 - 1) \\ &= (1-x^2)((x(1-x))^2 - 1). \end{aligned}$$

Note that

$$x(1-x) \leq \frac{(x+1-x)^2}{4} = \frac{1}{4}$$

holds. This yields

$$((x(1-x)) + 1)f(x) = (1-x^2)((x(1-x)) + 1).$$

If $x(1-x) + 1 \neq 0$, then it follows that

$$f(x) = 1 - x^2.$$

Claim — Let

$$\begin{aligned} \alpha &= \frac{1 + \sqrt{5}}{2}, \\ \beta &= \frac{1 - \sqrt{5}}{2} \end{aligned}$$

denote the roots of the polynomial $x^2 - x - 1$. Let λ denote a real number.

Let $g_\lambda: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g_\lambda(x) = \begin{cases} 1 - x^2 & \text{if } x^2 - x - 1 \neq 0, \\ \lambda & \text{if } x = \alpha, \\ 2\alpha - \alpha^4 - \alpha^2\lambda & \text{if } x = \beta. \end{cases}$$

Then g_λ satisfies the given functional equation.

Proof of the Claim. Since the roots of the polynomial $x^2 - x - 1$ add up to 1, it follows that for a real number t , both t and $1 - t$ are roots of the polynomial or neither of them is. Thus, if an element $t \in \mathbb{R}$ is not a root of the polynomial $x^2 - x - 1$, then

$$\begin{aligned} t^2 g_\lambda(t) + g_\lambda(1 - t) &= t^2(1 - t^2) + (1 - (1 - t)^2) \\ &= 2t - t^4. \end{aligned}$$

Then,

$$\begin{aligned} \alpha^2 g_\lambda(\alpha) + g_\lambda(1 - \alpha) &= \alpha^2 \lambda + g_\lambda(\beta) \\ &= \alpha^2 \lambda + (2\alpha - \alpha^4 - \alpha^2 \lambda) \\ &= 2\alpha - \alpha^4 \end{aligned}$$

holds. Moreover, we have

$$\begin{aligned} \beta^2 g_\lambda(\beta) + g_\lambda(1 - \beta) &= \beta^2(2\alpha - \alpha^4 - \alpha^2 \lambda) + g_\lambda(\alpha) \\ &= \beta^2(2\alpha - \alpha^4 - \alpha^2 \lambda) + \lambda \\ &= -2\beta - \alpha^2 \quad (\text{using } \alpha\beta = -1) \\ &= -2\beta - (1 - \beta)^2 \\ &= -2\beta - (1 - 2\beta + \beta^2) \\ &= -1 - \beta^2 \\ &= 2\beta - (1 + \beta)^2 \\ &= 2\beta - \beta^4. \end{aligned}$$

This completes the proof of the Claim. □

Thus, the functions satisfying the given functional equation are precisely the functions g_λ for real numbers λ . ■

Example 14.5. Determine all functions $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ satisfying

$$f(x) + 2f\left(\frac{1}{x}\right) = x$$

for all nonzero real numbers x .

Solution 89. Let $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ be a function satisfying the given functional equation. Let x be a nonzero real number. Note that

$$\begin{aligned} f(x) + 2f\left(\frac{1}{x}\right) &= x, \\ 2f(x) + f\left(\frac{1}{x}\right) &= \frac{1}{x} \end{aligned}$$

holds. Eliminating $f(1/x)$ from these two equations, we get

$$-3f(x) = x - \frac{2}{x},$$

which yields

$$f(x) = \frac{2}{3x} - \frac{x}{3} = \frac{2 - x^2}{3x}.$$

Moreover, the function $g: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ defined as

$$g(x) = \frac{2 - x^2}{3x}$$

for any nonzero real number x satisfies the given functional equation. Indeed, for any nonzero real number x , we have

$$\begin{aligned} g(x) + 2g\left(\frac{1}{x}\right) &= \frac{2 - x^2}{3x} + 2 \cdot \frac{2 - \frac{1}{x^2}}{3 \cdot \frac{1}{x}} \\ &= \frac{2 - x^2}{3x} + \frac{4x^2 - 2}{3x} \\ &= x. \end{aligned}$$

Thus, the only function satisfying the given functional equation is g . ■

Example 14.6. Determine all functions $f: \mathbb{R} \setminus \{0, 1\} \rightarrow \mathbb{R}$ such that

$$f(x) + 2f\left(\frac{1}{1-x}\right) + 3f\left(1 - \frac{1}{x}\right) = x$$

holds for all real numbers $x \neq 0, 1$.

Solution 90. Let $f: \mathbb{R} \setminus \{0, 1\} \rightarrow \mathbb{R}$ be a function satisfying the given functional equation. Let x be a real number such that $x \neq 0, 1$. Observe that $\frac{1}{1-x}$ and $1 - \frac{1}{x}$ are also real numbers different from 0 and 1. Note that

$$\begin{aligned} f(x) + 2f\left(\frac{1}{1-x}\right) + 3f\left(1 - \frac{1}{x}\right) &= x, \\ 3f(x) + f\left(\frac{1}{1-x}\right) + 2f\left(1 - \frac{1}{x}\right) &= \frac{1}{1-x}, \end{aligned}$$

$$2f(x) + 3f\left(\frac{1}{1-x}\right) + f\left(1 - \frac{1}{x}\right) = 1 - \frac{1}{x}$$

holds. This gives

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} f(x) \\ f\left(\frac{1}{1-x}\right) \\ f\left(1 - \frac{1}{x}\right) \end{pmatrix} = \begin{pmatrix} x \\ \frac{1}{1-x} \\ 1 - \frac{1}{x} \end{pmatrix}.$$

The inverse of the coefficient matrix is

$$\frac{1}{14} \begin{pmatrix} -5 & 4 & 3 \\ 3 & -5 & 4 \\ 4 & 3 & -5 \end{pmatrix}.$$

Thus, we have

$$\begin{pmatrix} f(x) \\ f\left(\frac{1}{1-x}\right) \\ f\left(1 - \frac{1}{x}\right) \end{pmatrix} = \frac{1}{18} \begin{pmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{pmatrix} \begin{pmatrix} x \\ \frac{1}{1-x} \\ 1 - \frac{1}{x} \end{pmatrix}.$$

In particular, we have

$$\begin{aligned} f(x) &= \frac{1}{18} \left(-5x + \frac{7}{1-x} + 1 - \frac{1}{x} \right) \\ &= \frac{-5x^2(1-x) + 7x - (x-1)^2}{18x(1-x)} \\ &= \frac{5x^3 - 5x^2 + 7x - x^2 + 2x - 1}{18x(1-x)} \\ &= \frac{5x^3 - 6x^2 + 9x - 1}{18x(1-x)}. \end{aligned}$$

Moreover, the function $g: \mathbb{R} \setminus \{0, 1\} \rightarrow \mathbb{R}$ defined as

$$g(x) = \frac{5x^3 - 6x^2 + 9x - 1}{18x(1-x)}$$

for any real number $x \neq 0, 1$ satisfies the given functional equation. Indeed, for any real number $x \neq 0, 1$, the real numbers $\frac{1}{1-x}$ and $1 - \frac{1}{x}$ are different from 0 and 1, and we have

$$\begin{aligned} g(x) + 2g\left(\frac{1}{1-x}\right) + 3g\left(1 - \frac{1}{x}\right) &= \frac{5x^3 - 6x^2 + 9x - 1}{18x(1-x)} + 2 \cdot \frac{5 - 6(1-x) + 9(1-x)^2 - (1-x)^3}{-18x(1-x)} \end{aligned}$$

$$\begin{aligned}
& + 3 \cdot \frac{5(x-1)^3 - 6x(x-1)^2 + 9x^2(x-1) - x^3}{-18x(1-x)} \\
& = \frac{1}{18x(1-x)} (5x^3 - 6x^2 + 9x - 1 \\
& \quad - 2(5 - 6 + 9 - 1 + 6x - 18x + 3x + 9x^2 - 3x^2 + x^3) \\
& \quad - 3(5x^3 - 6x^3 + 9x^3 - x^3 - 15x^2 + 12x^2 - 9x^2 + 15x - 6x - 5)) \\
& = \frac{1}{18x(1-x)} (5x^3 - 6x^2 + 9x - 1 - 2(x^3 + 6x^2 - 9x + 7) \\
& \quad - 3(7x^3 - 12x^2 + 9x - 5)) \\
& = \frac{18x^2 - 18x^3}{18x(1-x)} \\
& = x.
\end{aligned}$$

This shows that the only function satisfying the given functional equation is g . ■

§15 Functions

Exercise 15.1 (All-Russian Mathematical Olympiad 2014 Grade 10 Day 1 P2, AoPS, by O. Podlipsky). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(x)^2 \leq f(y)$$

holds for any $x, y \in \mathbb{R}$ with $x > y$. Show that

$$0 \leq f(x) \leq 1$$

holds for any $x \in \mathbb{R}$.

Walkthrough —

- (a) Show that $f(x) \geq 0$ for any $x \in \mathbb{R}$.
- (b) Using the given inequality, show that for any $x, y \in \mathbb{R}$ with $x > y$, and for any positive integer n , the inequality

$$f(x)^{2^n} \leq f(y)$$

holds.

- (c) Use the above to conclude that $f(x) \leq 1$ for any $x \in \mathbb{R}$.

Solution 91. Note that for any $x \in \mathbb{R}$, we have

$$f(x+1)^2 \leq f(x),$$

which shows that $f(x) \geq 0$.

To show that $f(x) \leq 1$ for any $x \in \mathbb{R}$, let us establish the following claim.

Claim — For any $x, y \in \mathbb{R}$ with $x > y$, and for any positive integer n , the inequality

$$f(x)^{2^n} \leq f(y)$$

holds.

Proof of the Claim. Note that the inequality holds for $n = 1$ by hypothesis. Let us assume that n is a positive integer such that the inequality holds for any $x, y \in \mathbb{R}$ with $x > y$. Then, for any $x, y \in \mathbb{R}$ with $x > y$, we have

$$x > \frac{x+y}{2} > y,$$

which implies that

$$f(x)^{2^n} \leq f\left(\frac{x+y}{2}\right) \quad \text{and} \quad f\left(\frac{x+y}{2}\right)^2 \leq f(y).$$

Combining these two inequalities and using that $f(x)$ is nonnegative, we obtain

$$f(x)^{2^{n+1}} \leq f(y).$$

By the principle of mathematical induction, the claim follows. □

Let x be an arbitrary real number. Using the above claim, we obtain

$$f(x)^{2^n} \leq f(x-1)$$

for **any** positive integer n . If $f(x) \leq 1$, then we are done. It remains to consider the case that $f(x) > 1$, which we assume from now on. It follows that

$$f(x-1) \geq 1 + 2^n (f(x) - 1)$$

for **any** positive integer n . This is impossible since $f(x) - 1$ is positive.

Combining the above, we conclude that

$$0 \leq f(x) \leq 1$$

holds for any $x \in \mathbb{R}$. ■

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